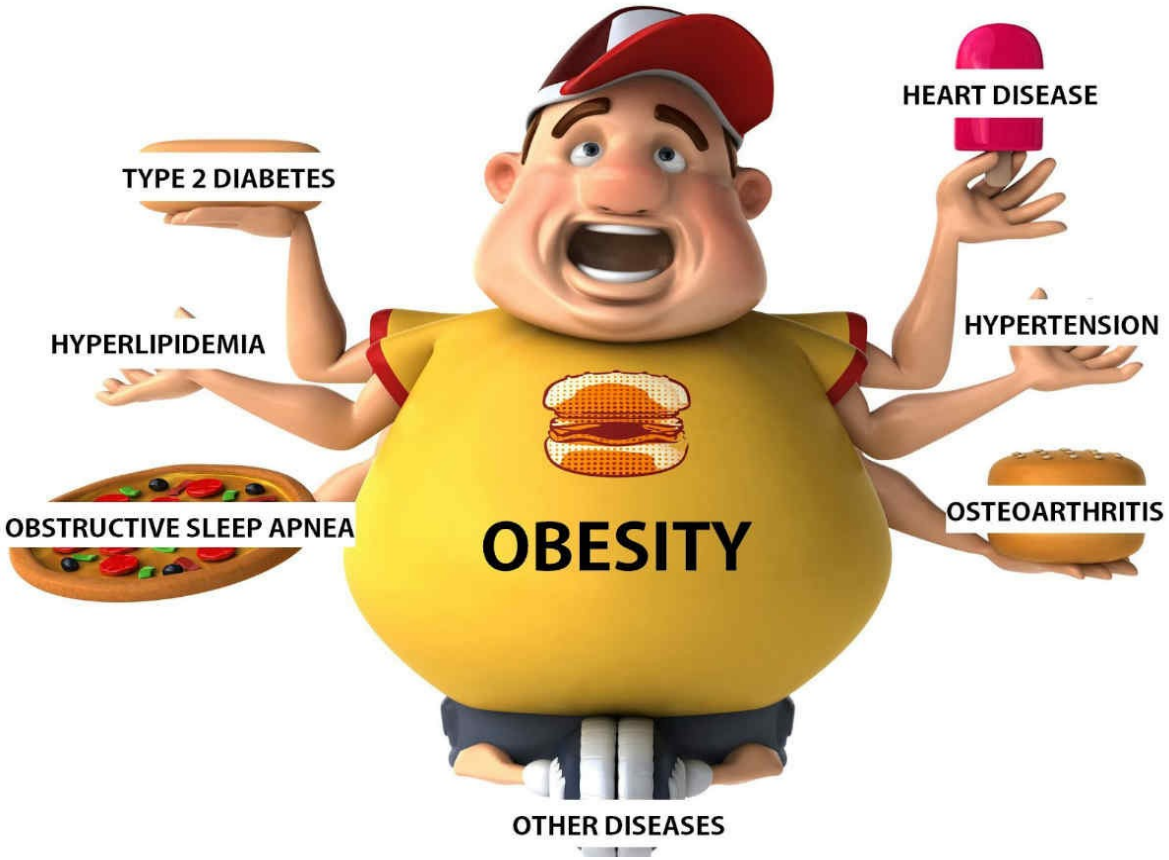
Title of the lecture:

LIPID METABOLISM



Professor

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1-Introduction:

Lipids are a heterogeneous group of water-insoluble, bio-organic molecules. Because of their insolubility in aqueous solution, body lipids are generally subdivided into:

- **Membrane-associated lipids**
- **Droplets of triglycerides**
- **Transport in plasma in association with protein**

2-Biomedical importance

Not surprisingly, deficiency or imbalance of lipid metabolism can lead to several critical clinical disorders observed by physicians, such as:

- ✓ **Atherosclerosis &**
- ✓ **Cardiovascular disease**
- ✓ **Obesity &**
- ✓ **Truncal obesity**
- ✓ **Ketosis**



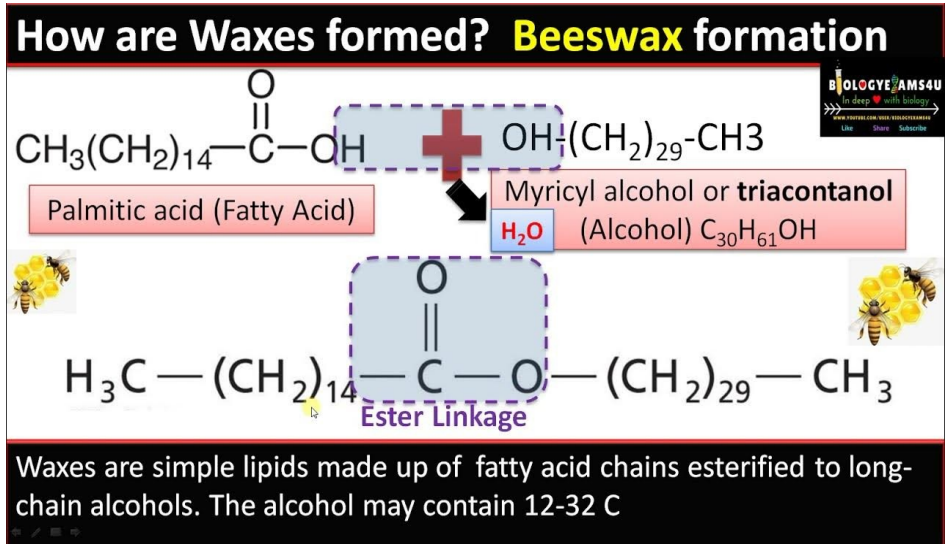
3-Functions of lipids:

- Lipids are a highly efficient form of energy storage (1gr. = 9.1 calories)
- Lipids are integral components of cell membranes
- Lipids serve as thermal insulators in the subcutaneous tissues & around certain organs
- Metabolic regulations,
- They act as electrical insulators in neurons.
- Lipids aid in the absorption of fat-soluble Vit.
- Lipids supply so-called essential fatty acids.

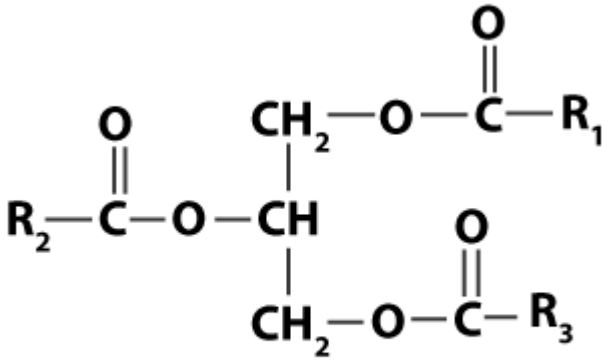
4-Classification of lipids

a. Simple lipids: These are lipids composed of fatty acids esterified with alcohols. They include:

- **Neutral Fats: Formed by the esterification of fatty acids with glycerol. (Triglyceride=TGs)**
- **Waxes: Formed by the esterification of fatty acids with long-chain (high molecular weight) alcohols other than glycerol**



What is TG composed of?



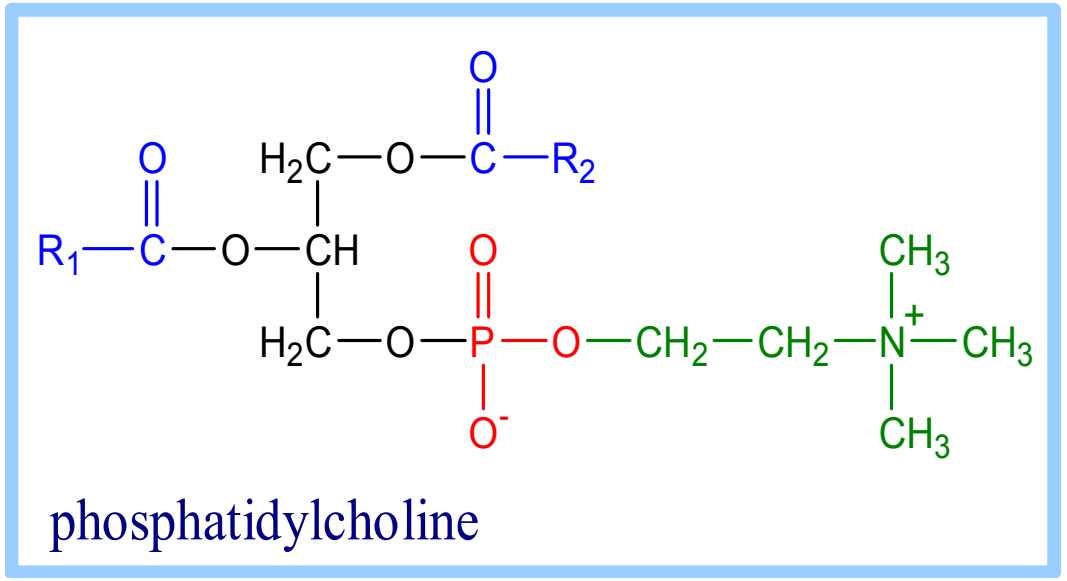
Triglyceride

b. Complex lipids (Compound lipids):

These are lipids in which fatty acids are esterified with an alcohol backbone, and in addition, they contain other functional groups such as:

Phospholipids such as:

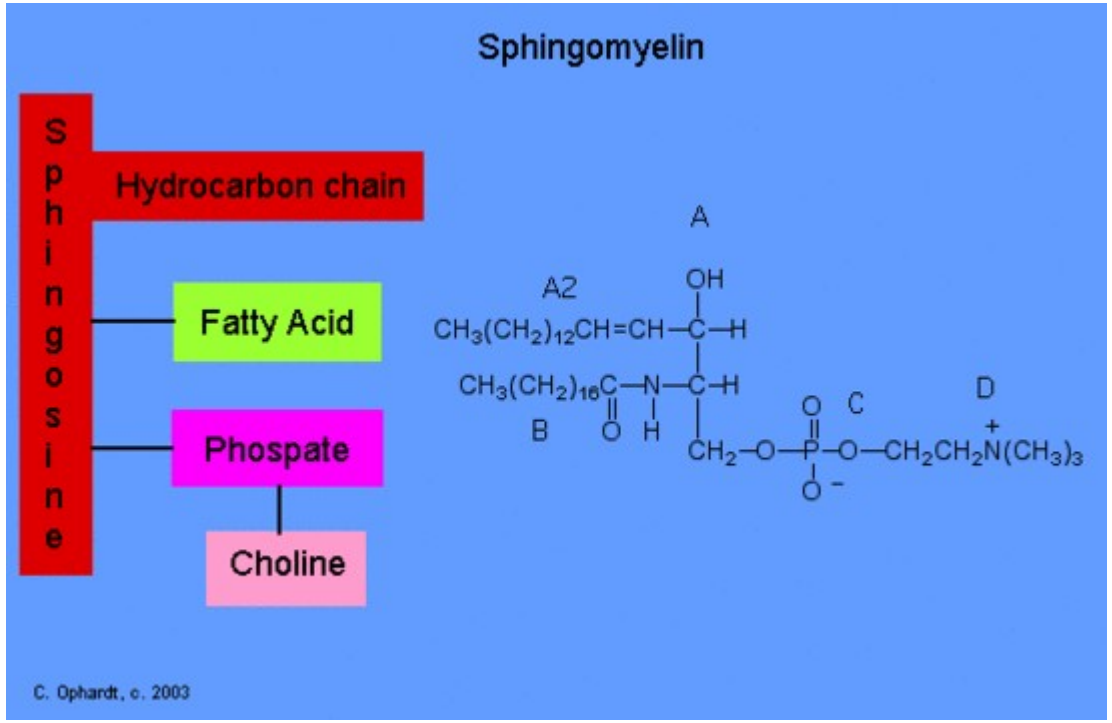
- ✓ **Phosphatidyl Cephalin**
- ✓ **Phosphatidyl Serines**
- ✓ **Phosphatidyl Choline (lecithin)**
- ✓ **Phosphatidyl Inositol**
- ✓ **Sphingomyelin**



phosphatidylcholine

Sphingomyelin:

Is a type of sphingolipid, which is a class of lipids found in the membranes of animal cells. It is particularly abundant in the myelin sheath that surrounds and insulates nerve fibers in the central and peripheral nervous systems.



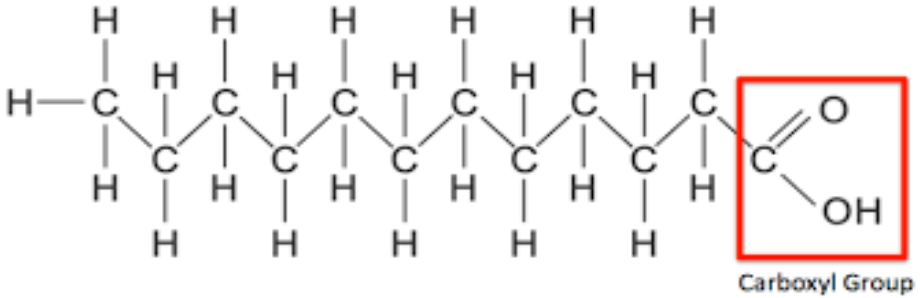
Non-phosphorylated lipids such as:

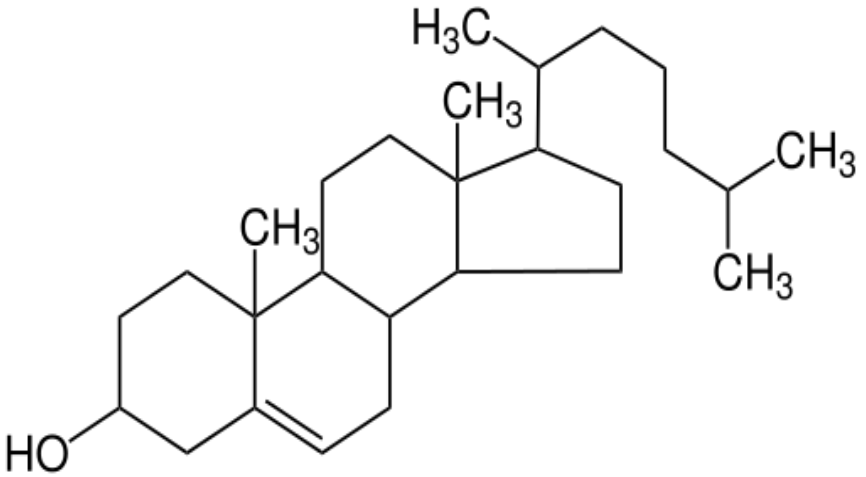
- **Glycolipids: e.g, cerebrosides, which are lipids containing both a sugar & a ceramide (sphingosine + fatty acid) component. They are important components of nerve cell membranes.**
- **Sulfolipids, e.g., Sulfatide, which are a major component in the nervous system & are found in high levels in the myelin sheath.**
- **Lipids complexed to other compounds, like lipoproteins**

c. Precursor & Derived lipids:

A substance derived from the above groups by hydrolysis like

- ✓ **Fatty acids,**
- ✓ **Steroids,**
- ✓ **Prostaglandins &**
- ✓ **Ketone bodies.**
- ✓ **Cholesterol**





5-Digestion & absorption of dietary lipids

Digestion of the lipids can be divided into:

a)Processing of dietary lipids in the stomach

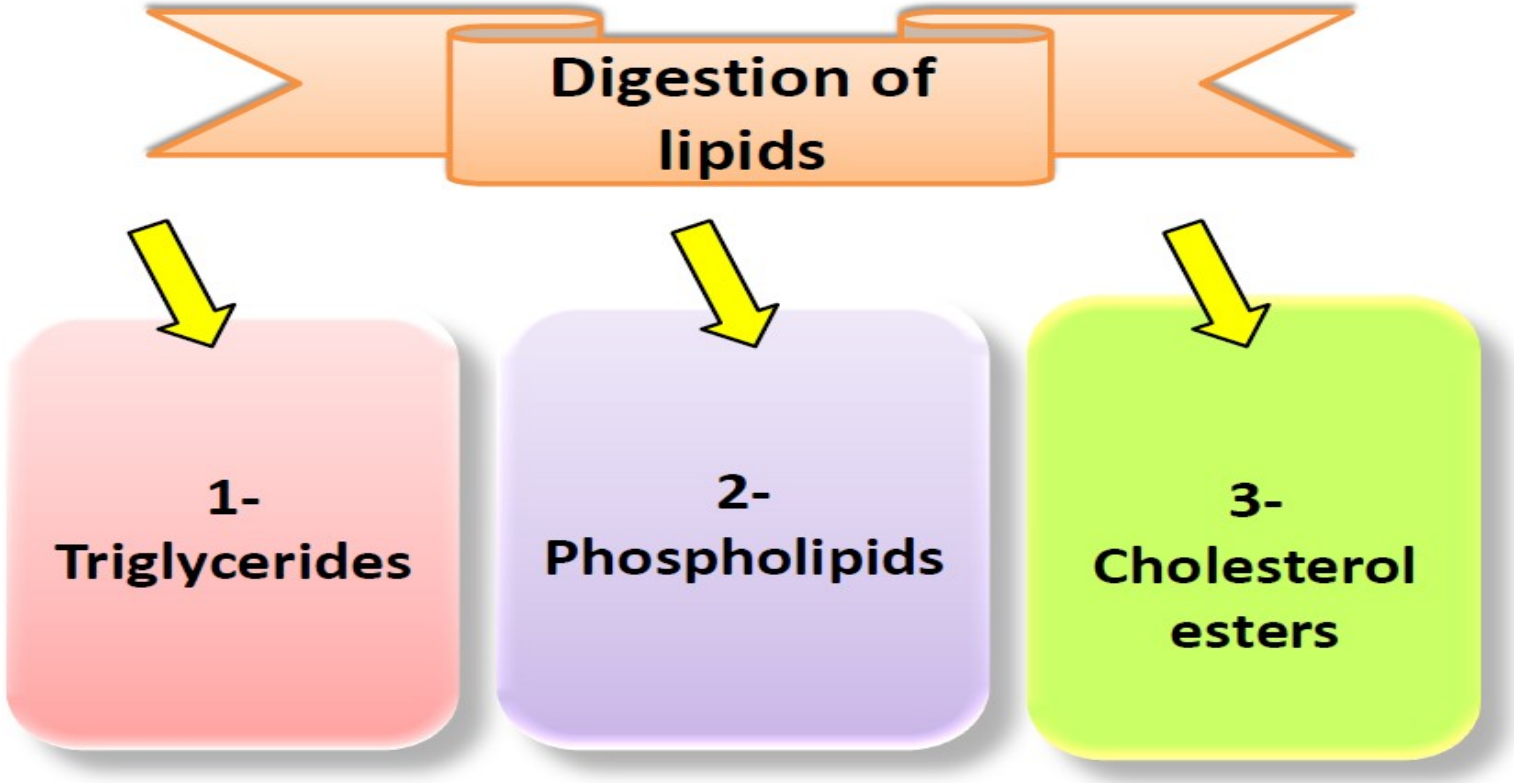
b)Emulsification of dietary lipids in the small intestine

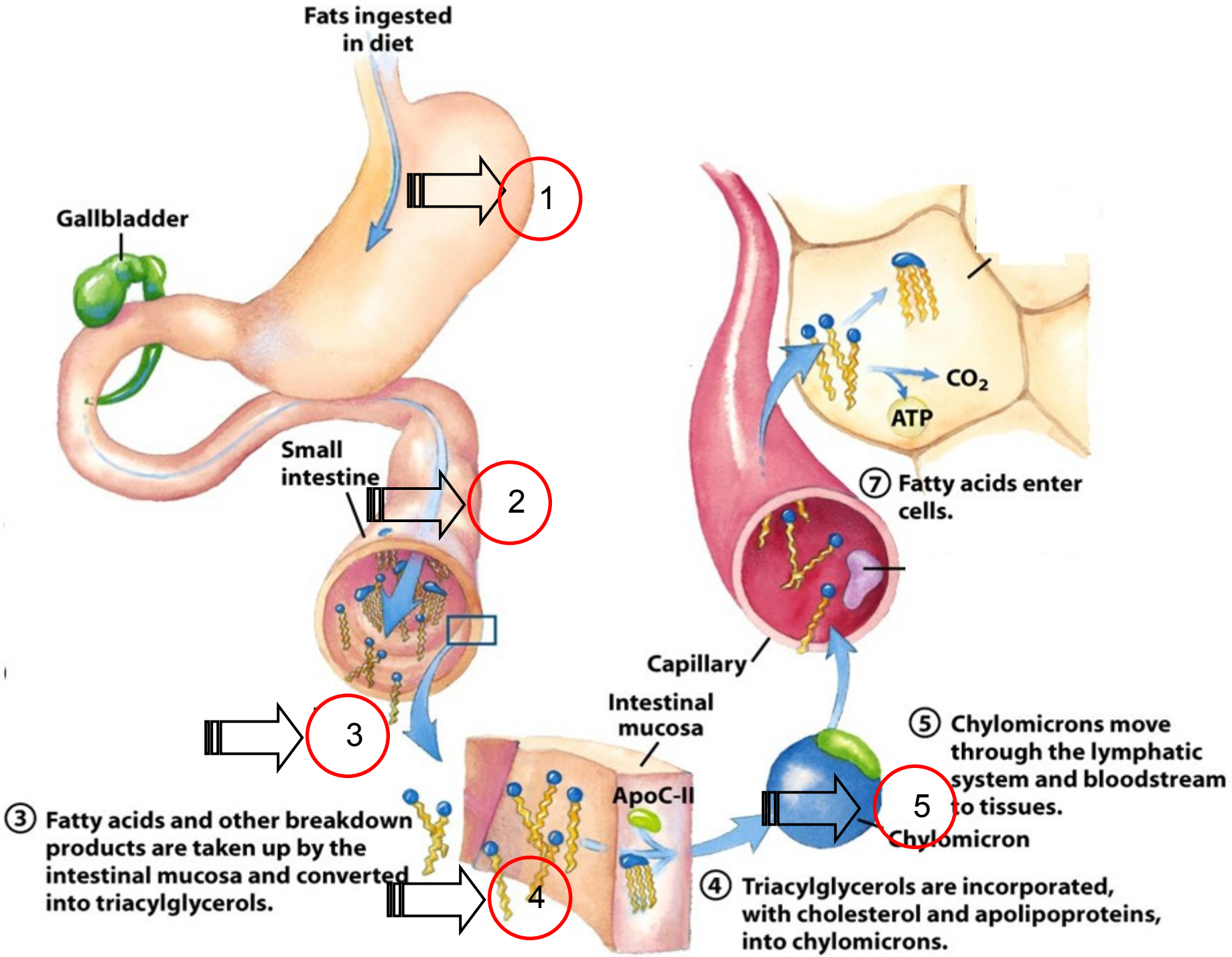
c)Degradation of dietary lipids by pancreatic enzymes

d)Absorption of the dietary lipids

Note:

An adult ingests about 60-150 gr,. of which more than 90% is normal TGs, the remaining consists primarily of Cholesterol, Cholesterol-ester, phospholipids, & non-esterified FAs.

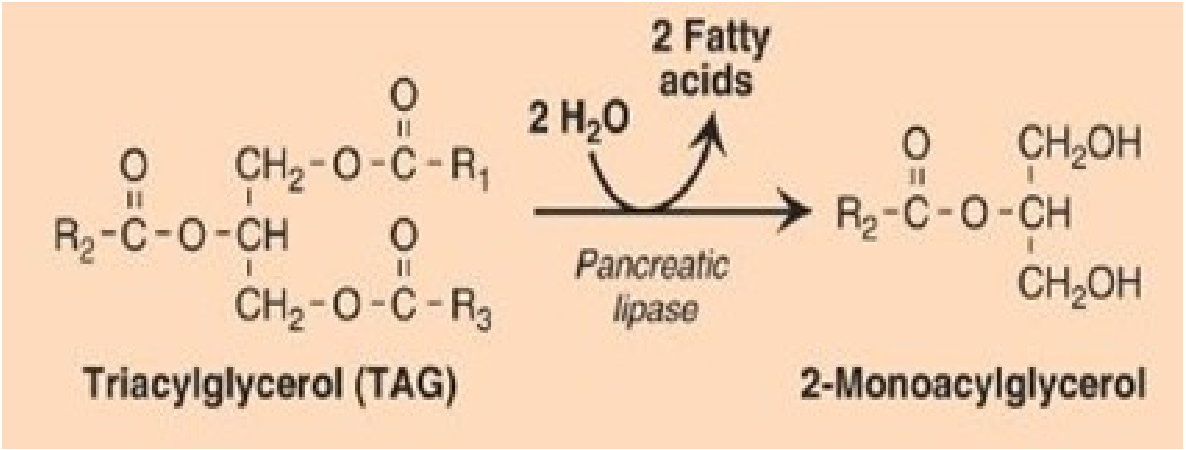



Fats ingested in diet
Gallbladder
1
Small intestine
2
3
Intestinal mucosa
ApoC-II
4
5
Chylomicron
Capillary
CO_2
ATP
7 Fatty acids enter cells.
5 Chylomicrons move through the lymphatic system and bloodstream to tissues.
3 Fatty acids and other breakdown products are taken up by the intestinal mucosa and converted into triacylglycerols.
4 Triacylglycerols are incorporated, with cholesterol and apolipoproteins, into chylomicrons.

A- Processing of dietary lipids in the stomach:

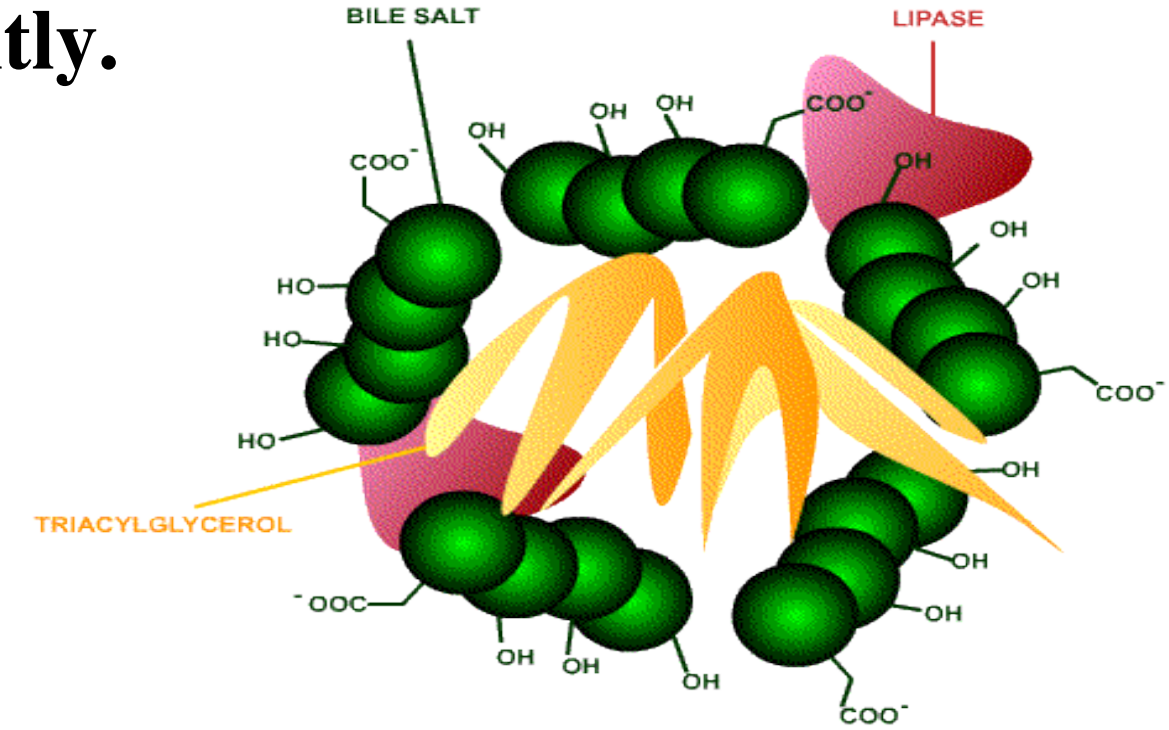
The digestion of lipids begins in the stomach and is catalyzed by lingual lipase (which originates from glands at the back of the tongue)

TGs —(**Lingual & gastric lipases**, **<12 C**)→ **2-Monoacyl glycerol + 2FA**



B-Emulsification of dietary lipids in the small intestine:

- **Emulsification, facilitated by bile salts, breaks down large, hydrophobic lipid droplets into smaller ones, significantly increasing their surface area.**
- **This process allows digestive enzymes, which function at the interface between lipid droplets and the surrounding aqueous solution, to work more efficiently.**



C- Degradation of dietary lipids by pancreatic enzymes

TG —(Lipase)→ **2-Mono acyl glycerol + FFAs**

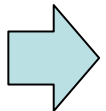
Cholesterol ester —(Cholesterol ester hydrolase (Esterase))→ **Ch. + FFA**

Phospholipids —(Ph. Lipase A_2)→ **Lysophospholipids + FA**

It removes one FA from C_2

Lysophospholipids —(Lysophospholipase)→ **Glyceryl phosphoryl base + FA**

Removes FA from C_1



D- Absorption of lipids

- FAs, glycerol, and other products of lipid digestion are absorbed by the intestinal mucosal cells (enterocytes).
- FAs and glycerol are reassembled (Reconstructed) within these cells into triglycerides (TGs). These triglycerides are then combined with cholesterol, cholesterol esters, and apolipoproteins to form chylomicrons,

What is a chylomicron?

6-Control of lipid digestion:

The pancreatic secretion of hydrolytic enzymes in the small intestine is hormonally controlled by

Cholecystokinin (Pancreozymin):

It is a peptide hormone of the GI system that plays a key role in facilitating digestion within the small intestine.

It acts on the

- **Gallbladder: it causes the gallbladder to constrict and release bile acids**
- **Exocrine cells of the pancreas cause them to release digestive enzymes.**
- **It also decreases gastric motility, resulting in a slower release of gastric contents into the small intestine,**

7- Plasma lipids & lipoproteins

A-Plasma lipids:

These are lipids present in the circulating blood, primarily in the form of:

- **TGs,**
- **Cholesterol,**
- **Phospholipids, and**
- **Free FAs.**

These lipids are transported in the blood by binding to lipoproteins, such as

- **Chylomicrons,**
- **Very low-density lipoproteins (VLDL),**
- **Low-density lipoproteins (LDL), &**
- **High-density lipoproteins (HDL)**

Plasma lipids make up a very small percentage of the total lipids in the body compared to the vast lipid stores in

- **Adipose tissue,**
- **Cell membranes, and**
- **Other organs.**

Despite their small proportion,

- **They play a vital role in metabolic processes.**
- **They also act as markers of lipid metabolism.**

Variations in their levels may reflect metabolic conditions such as fasting, feeding, or disorders like hyperlipidemia, diabetes, & cardiovascular diseases

Constituents & amounts of plasma lipids

The principle plasma lipids, their range, and average value

Total lipids	**500mg/dl (400-700)**
TGs	**100mg/dl (10-160)**
PLs	**200mg/dl (150-250)**
Total Cholesterol	**200mg/dl (150-200)**
Chol. ester	**145mg/dl (90-200)**
Free Chol.	**55mg/dl (40-70)**
NEFA(Non-esterified fatty acids)	**5-20mg/dl**

B- Lipoproteins:

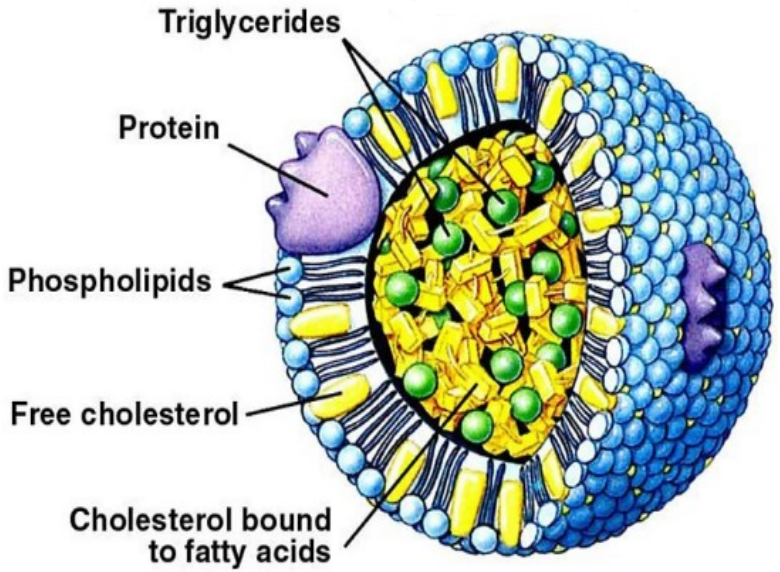
They are the primary transport form of lipids in plasma.

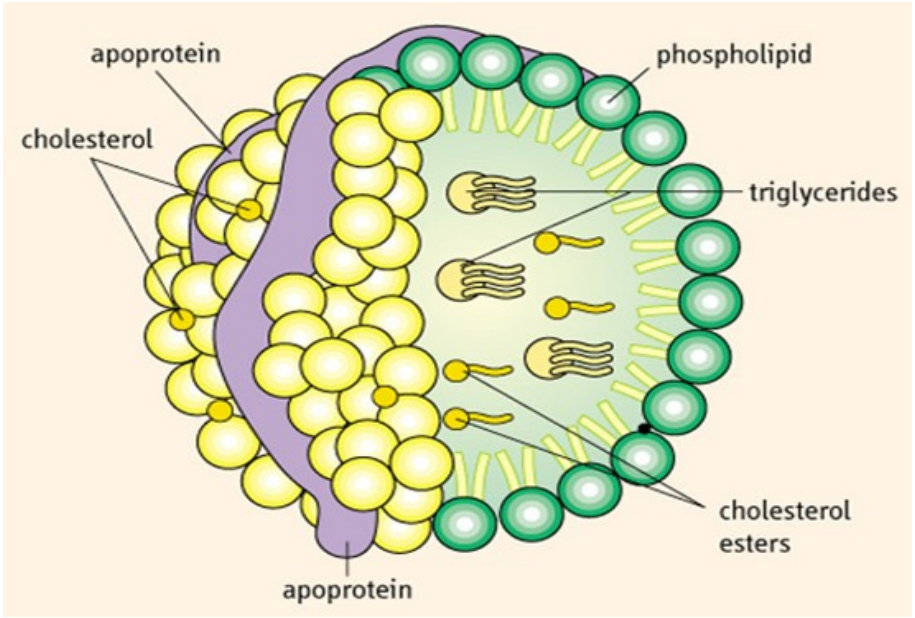
Note:

- **Since lipids are hydrophobic and insoluble in water, how are they transported through the bloodstream?**
- **What makes lipids soluble?**
- **This is solved by combining the insoluble lipids with proteins to form a hydrophilic lipoprotein complex.**

Structure of lipoproteins:

- **Lipoprotein has a lipid core of non-polar TGs, Ch.E surrounded by more polar phospholipids, Ch. and proteins**
- **The protein moiety is named apolipoproteins or apoproteins (Apo).**





8-Classification of plasma lipoproteins

Lipoproteins can be divided into 4 types by electrophoresis and ultracentrifugation:

- **Chylomicrons, CM**
- **Very low-density LP, VLDL**
- **Low-density Lp, LDL**
- **High-density Lp, HDL**

9-Functions of Apoproteins (Apo)

➢ Structural stabilization:

Stabilize & maintain the molecular structure of lipoproteins

➢ Enzyme activation:

Certain apolipoproteins serve as cofactors for enzymes involved in lipid metabolism.

➢ Recognition and binding:

Apolipoproteins contain specific binding sites that allow them to interact with cell surface receptors, facilitating the uptake & metabolism of lipoprotein particles by various tissues.

Apoprotein Apo	Functions
B-100	✓ Secretion of VLDL from liver ✓ Ligand for LDL receptor
B-48	✓ Binds LDL receptor ✓ Secretion of chylomicrons from intestine
E	✓ Binds LDL receptor
A-I	✓ Major structral protein of HDL ✓ LCAT activator (lecithin: cholesterol acyl transferase
A-II	✓ Unknown
C-I	✓ Inhibits the activity of cholesteryl ester transfer protein
C-II	✓ Avtivator of LPL
C-III	✓ Inhibitor of LPL activity

10-Lipid Metabolism

1- Lipoprotein metabolism

- ✓ Chylomicrons
- ✓ VLDLs Very Low-Density Lipoproteins
- ✓ IDLs Intermediate Density Lipoproteins
- ✓ LDLs Low-Density Lipoproteins
- ✓ HDLs High-Density Lipoproteins

2- Lipid metabolism

Where …… is formed

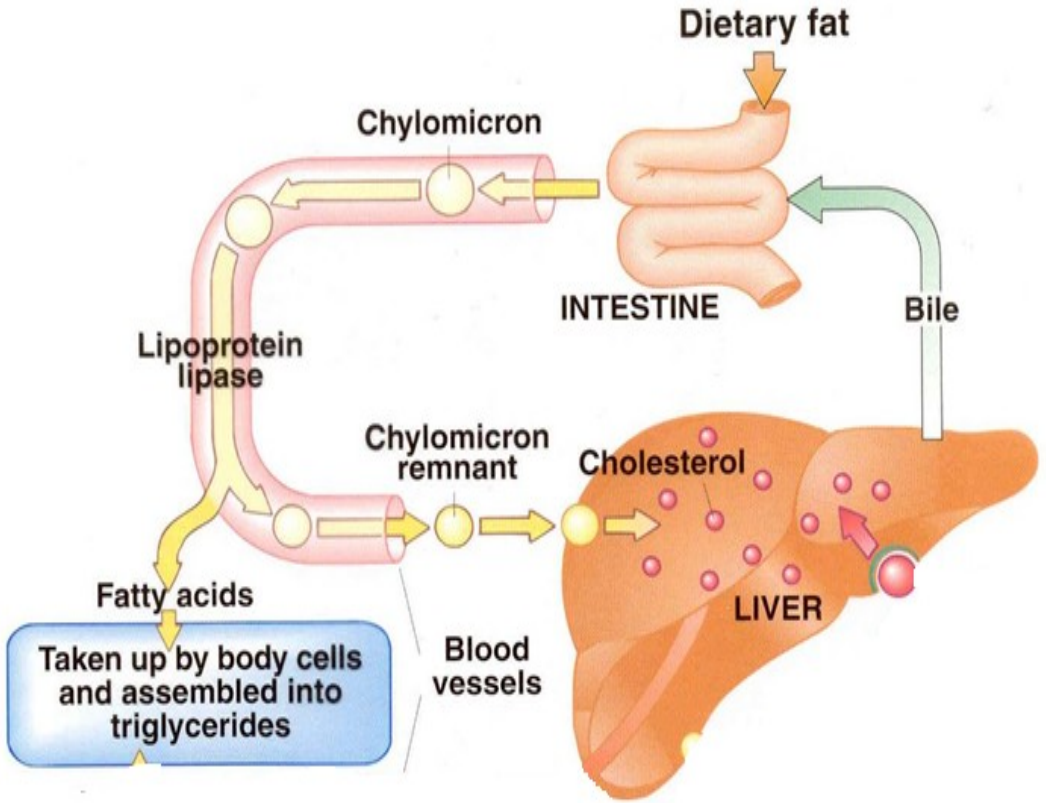
What is the composition & function of ……???

10.1-Chylomicrons

Origin: Mucosal cells of the intestine.

Major lipid: TGs

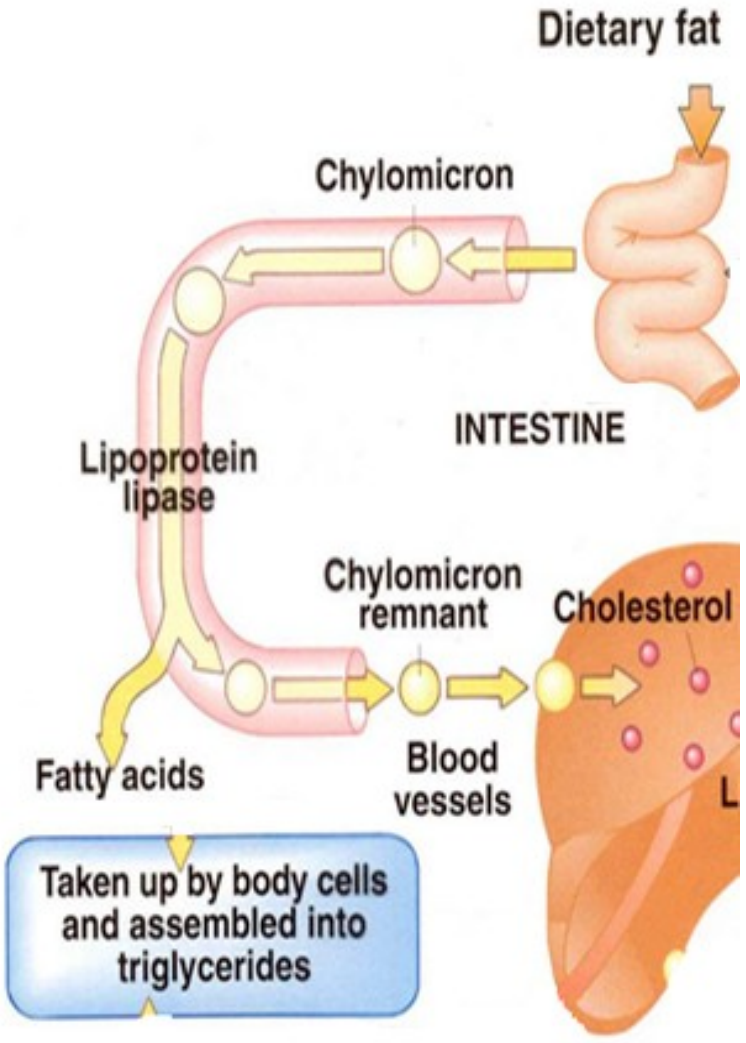
Function: Transport exogenous lipids from the intestine to other parts of the body.



Chylomicron Metabolism via Lipoprotein Lipase

LPL-Mediated Hydrolysis (Where it happens?):

- In the bloodstream (capillaries of AT, muscle, & the heart), chylomicrons interact with LPL, which is activated by Apo C-II,
- TGs within the chylomicrons are broken down by LPL, releasing free FFAs that are utilized by muscle and AT.
- This process results in the formation of chylomicron remnants, Chyl. remnants are then taken up by the liver



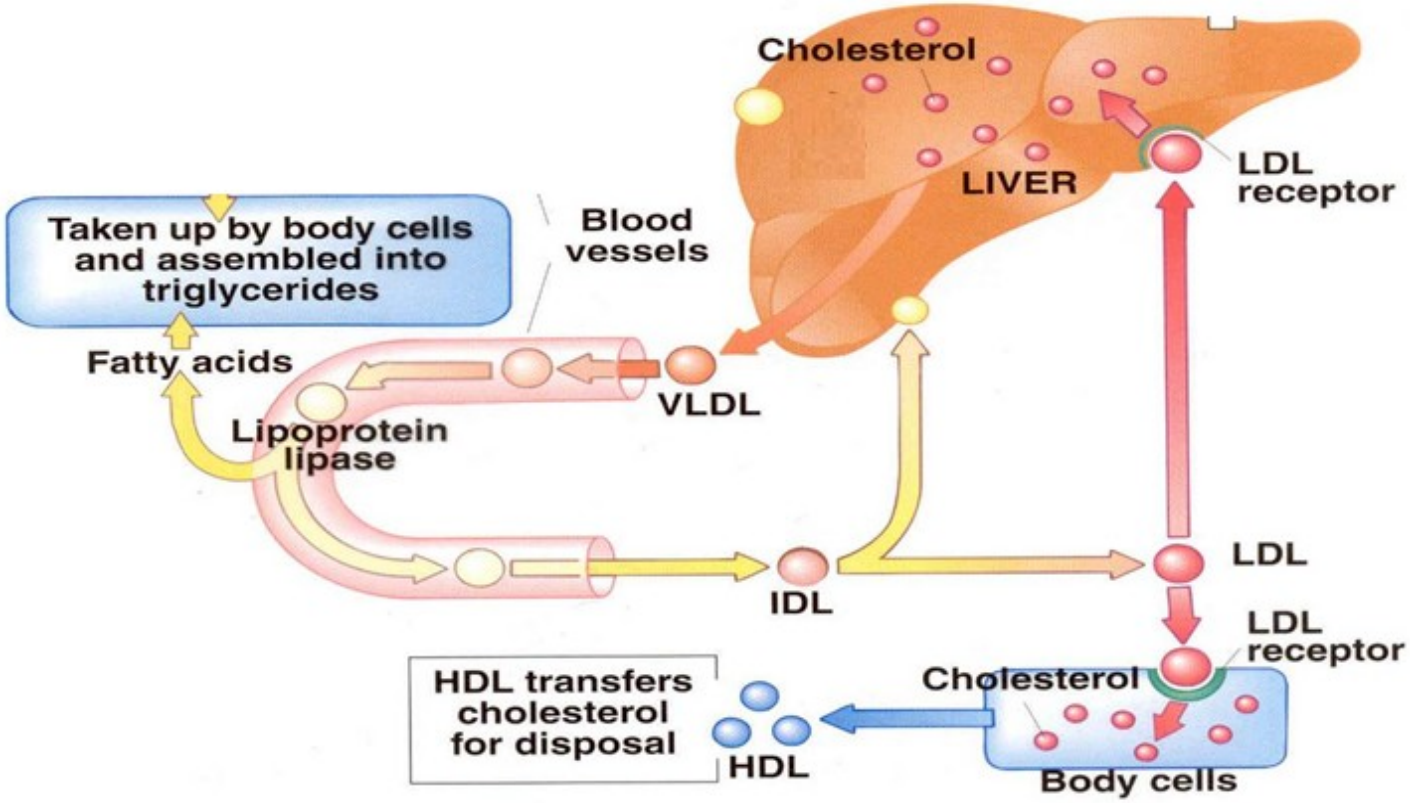
Note: Lipoprotein lipase (LPL) – an enzyme that breaks down triglycerides into free fatty acids (FFAs).

10.2-VLDLs

Origin: Hepatic cells.

Major lipid: TG

Function: Transport endogenous lipids from the liver to extrahepatic tissues.

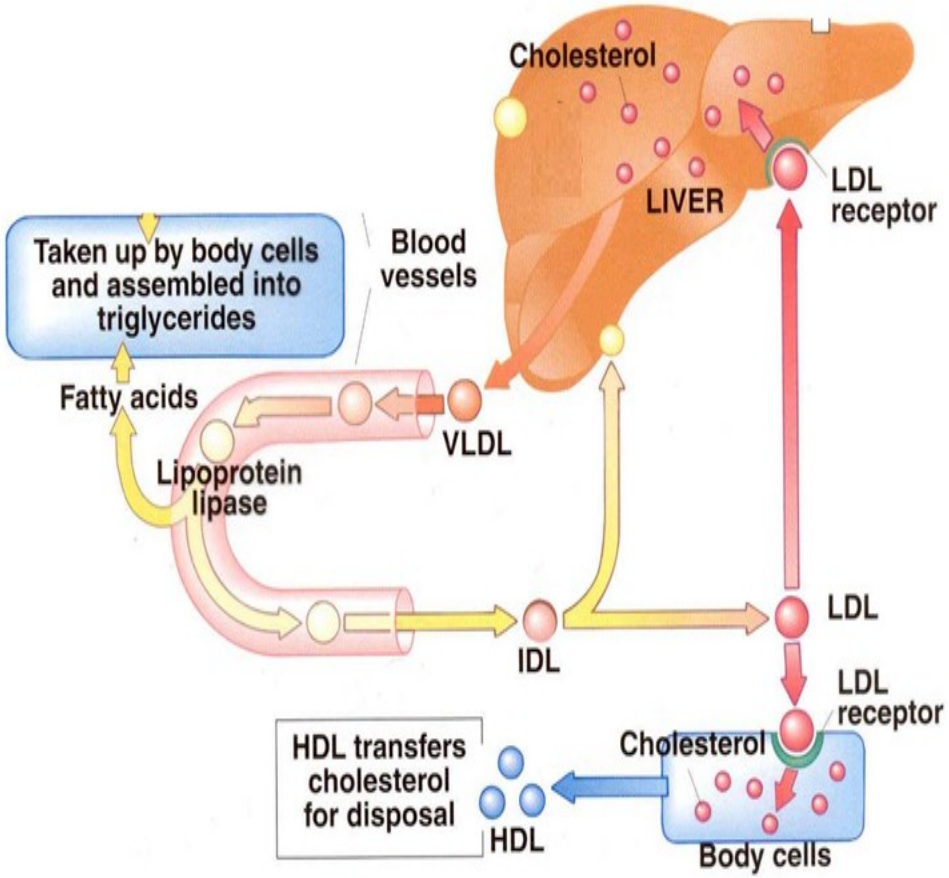


VLDL Metabolism via Lipoprotein Lipase

Reactions:

LPL-Mediated Hydrolysis:

- Under the LPL catalyst, TG is hydrolyzed to form FAs & glycerol.
- VLDL is converted to IDL and has an equal amount of cholesterol & TG.
- 50% of IDL is captured by the liver and half of it further undergoes the same procedure as VLDL does.
- The last remnant is LDL.



10.3- LDLs (Bad cholesterol)

Origin: VLDL

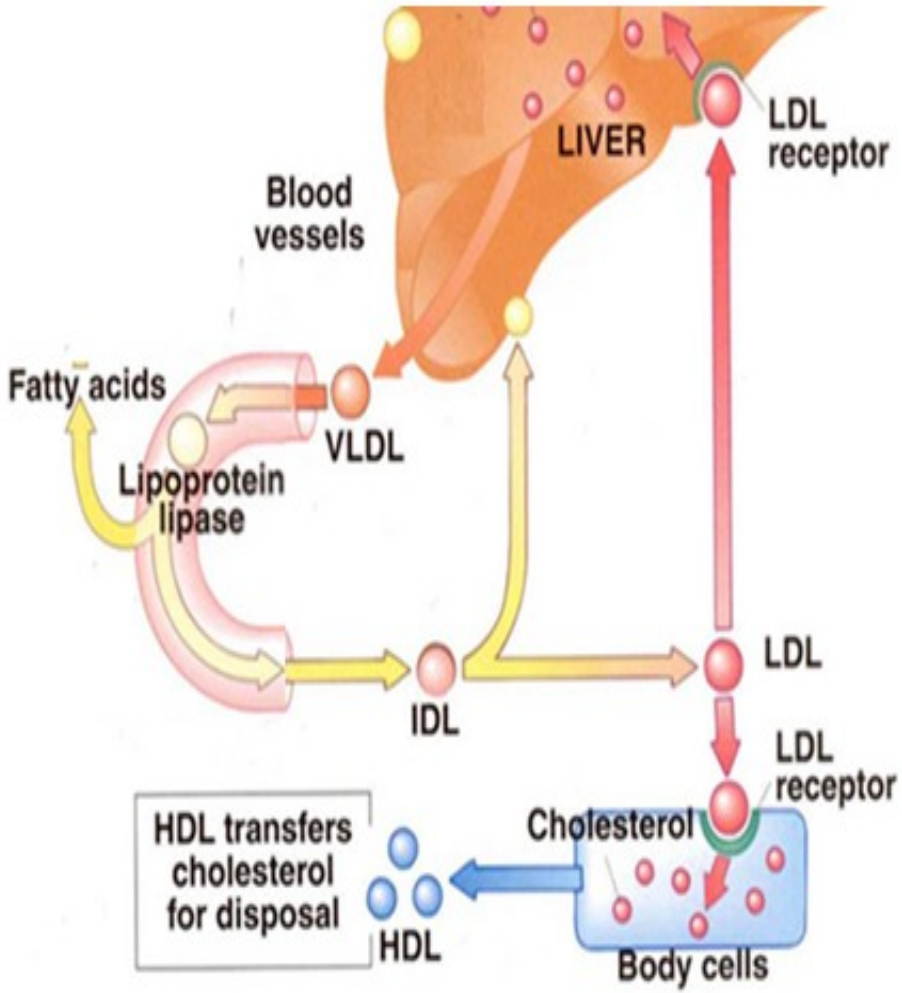
Major lipids: Cholesterol

Reaction:

50% of LDL is degraded in the liver; 50% of LDL is captured by extrahepatic tissues. These include tissues that require cholesterol for various functions

Function:

Transport endogenous lipids from the liver to extrahepatic tissues.



Functions of LDL in the cells:

Here are the functions of LDL in cells:

- **Steroid hormone & Vitamin D synthesis**
- **Bile acid production**
- **Intracellular signaling**
- **Lipoprotein and cell membrane synthesis**
- **Cholesterol metabolism:**
  - ✓ **Inhibit HMG-CoA reductase to inhibit Ch synthesis.**
  - ✓ **Inhibit biosy. of the LDL receptor at the transcription level.**
  - ✓ **Stimulate acyl CoA-cholesterol acyltransferase (ACAT) activity, stimulate the formation of Ch-E from free Ch & accumulate in the plasma.**

10.4- HDLs (Good cholesterol)

Origin: liver (major), intestine, Chylomicron & VLDL.

Major lipids: Cholesterol ester & phospholipids.

Reaction:

- **One of the key enzymes involved in HDL function is Lecithin-Cholesterol Acyltransferase (LCAT).**
- **LCAT is an enzyme that catalyzes the conversion of free cholesterol to cholesteryl esters.**
- **It uses lecithin (phosphatidylcholine, a component of HDL) to transfer fatty acids to free cholesterol.**

Role of LCAT in Lipid Metabolism:

- **Cholesterol Transport**
- **HDL Maturation**
- **Prevention of Atherosclerosis**

:

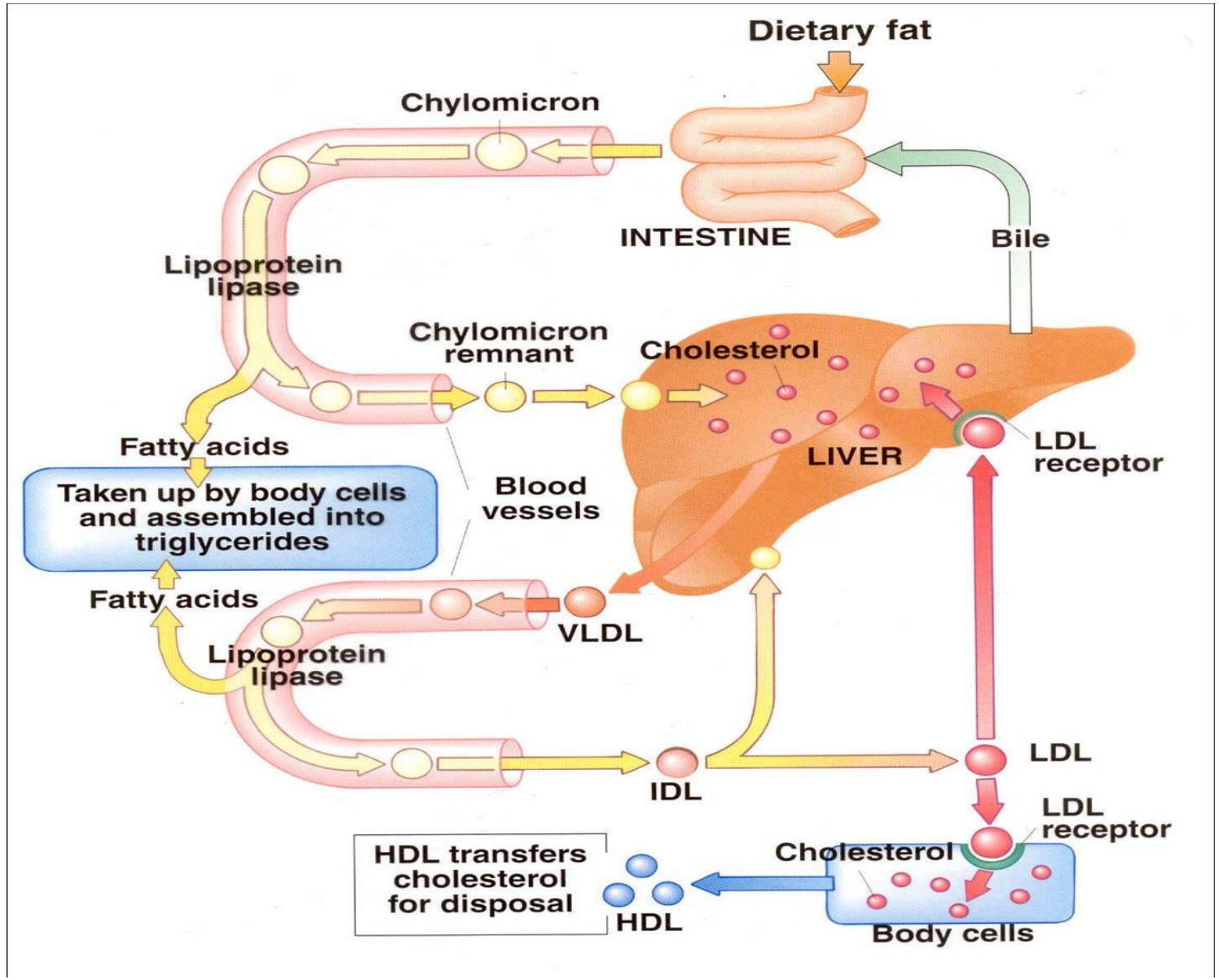
Function:

1- Reservoir for apo C-II & E

2- Uptake of unesterified Ch

3- Esterification of cholesterol

4-Transport Ch. from peripheral tissues to the liver. The liver then flushes it from the body by converting the cholesterol to bile salts


Dietary fat
Chylomicron
INTESTINE
Bile
Lipoprotein lipase
Chylomicron remnant
Cholesterol
LIVER
LDL receptor
Fatty acids
Taken up by body cells and assembled into triglycerides
Blood vessels
Fatty acids
VLDL
Lipoprotein lipase
IDL
LDL
LDL receptor
Cholesterol
HDL transfers cholesterol for disposal
HDL
Body cells

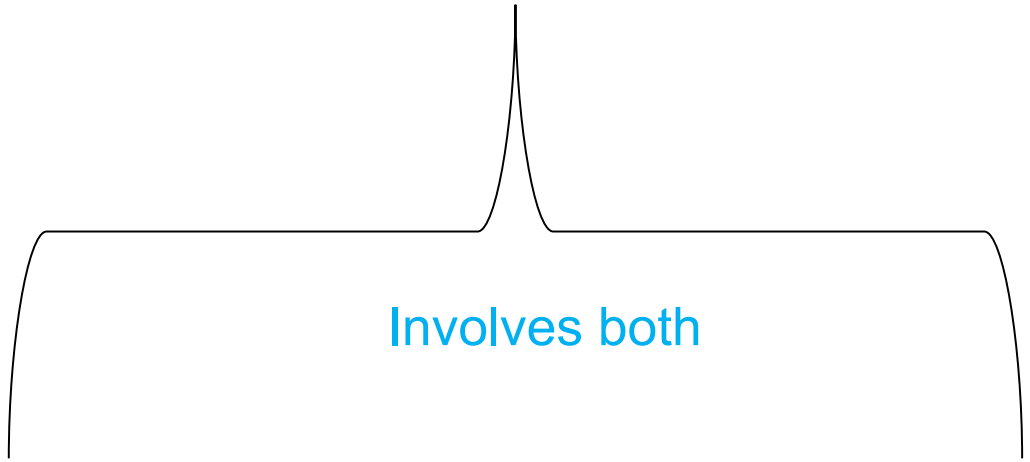
11- Lipids metabolism:

11.A -TGs

a. Esterification (Biosynthesis)
b. Lipolysis (Breakdown)

11.B -Cholesterol

11.A- Triglyceride Metabolism



a-Esterification

(Biosynthesis)

b-Lipolysis

(Breakdown)

Catabolism & anabolism are not the forward & reverse processes of the same; they are entirely different pathways involving different reactants & enzymes

A-Esterification (Biosynthesis) of TGs:

For TG biosynthesis, 2 substrates are required

- Fatty acids (Activated Fatty-Acyl Co-A).
- Glycerol (α-Glycerol phosphate.

What are the sources of

- Glycerol phosphate?
- Acyl-CoA?

How is α-glycerol phosphate formed in the body?

I-Source of α-glycerol phosphate:

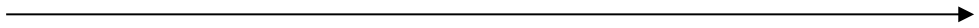
Mainly 2 sources

1. Conversion of glycerol to α-glycerol-P by gly. kinase in the presence of ATP
2. Glucose oxidation: dihydroxyacetone phosphate is converted to α–gly.-P without ATP

II- Substrate activation: Conversion of FFAs to their activated form

FFA —(Thiokinase / CoA Synthetase)→ acyl-CoA

Note: CoA: Its primary function is to carry and transfer acyl groups in various biochemical reactions.



Can AT utilize the glycerol formed by lipolysis in the synthesis of TG?

- **Glycerol kinase is absent in A.T**
- **Glycerol produced by lipolysis in AT can not be utilized to provide α-gly-P in AT.**
- **Glycerol from the liver, kidney & other tissues that possess Glycerol Kinase is utilized for α-glycerol phosphate synthesis.**

A-Triglyceride Biosynthesis Steps:

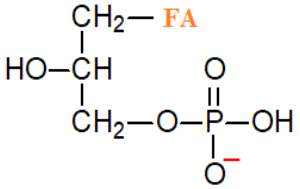
A.Gly-3-P & fatty acyl-CoA molecules undergo esterification, resulting in the formation of lysophosphatidic acid. This step is catalyzed by glycerol-3-phosphate acyltransferase.

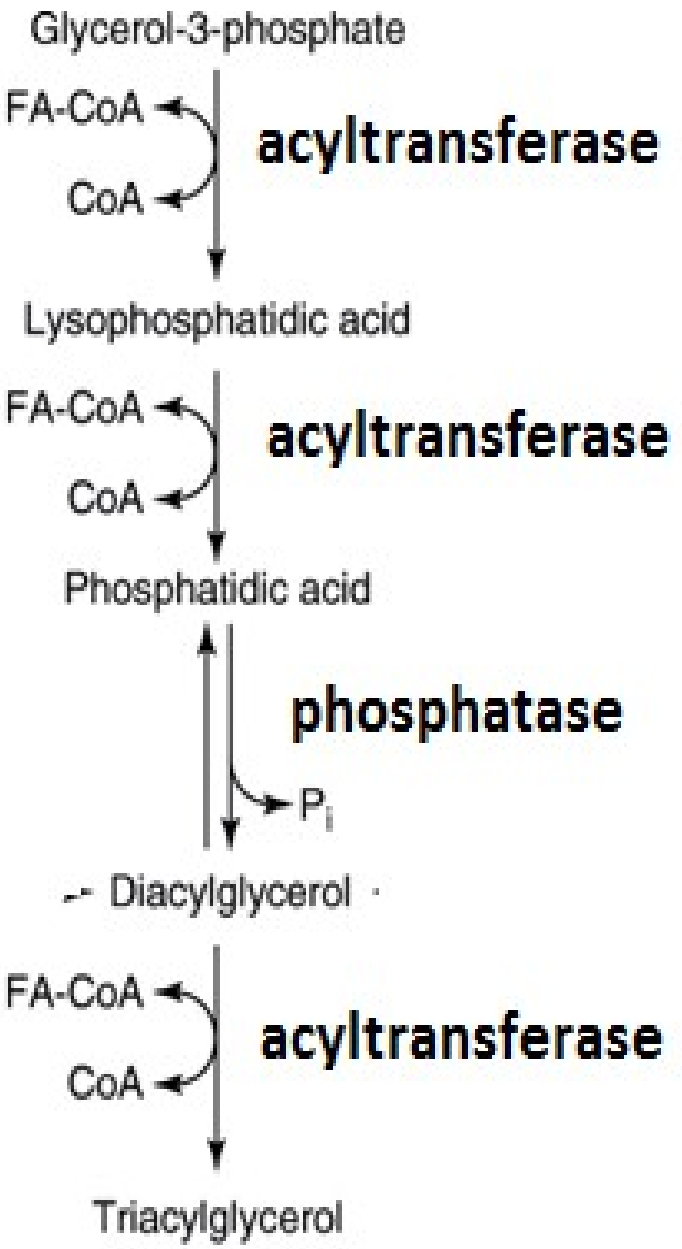
B.Lysophosphatidic acid & fatty acyl-CoA molecules undergo esterification, resulting in the formation of phosphatidic acid. This step is catalyzed by lysophosphatidic acid acyl transferase

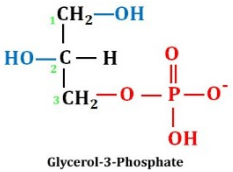
C. Phosphatidic acid undergoes dephosphorylation. Phosphatase catalyzes this step

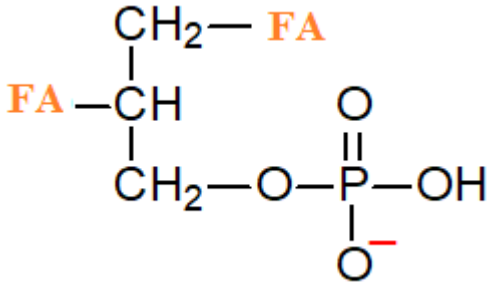
D. Diacyl glycerol & fatty acyl-CoA molecules undergo esterification, resulting in the formation of TG, This step is catalyzed by acyl transferase

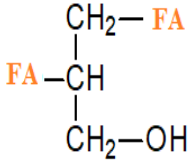
Gly-3-P and fatty acyl-CoA molecules undergo esterification, resulting in the formation of lysophosphatidic acid, This step is catalyzed by glycerol-3-phosphate acyltransferase

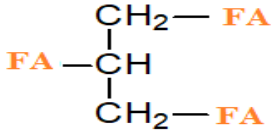












An overview of TG biosynthesis

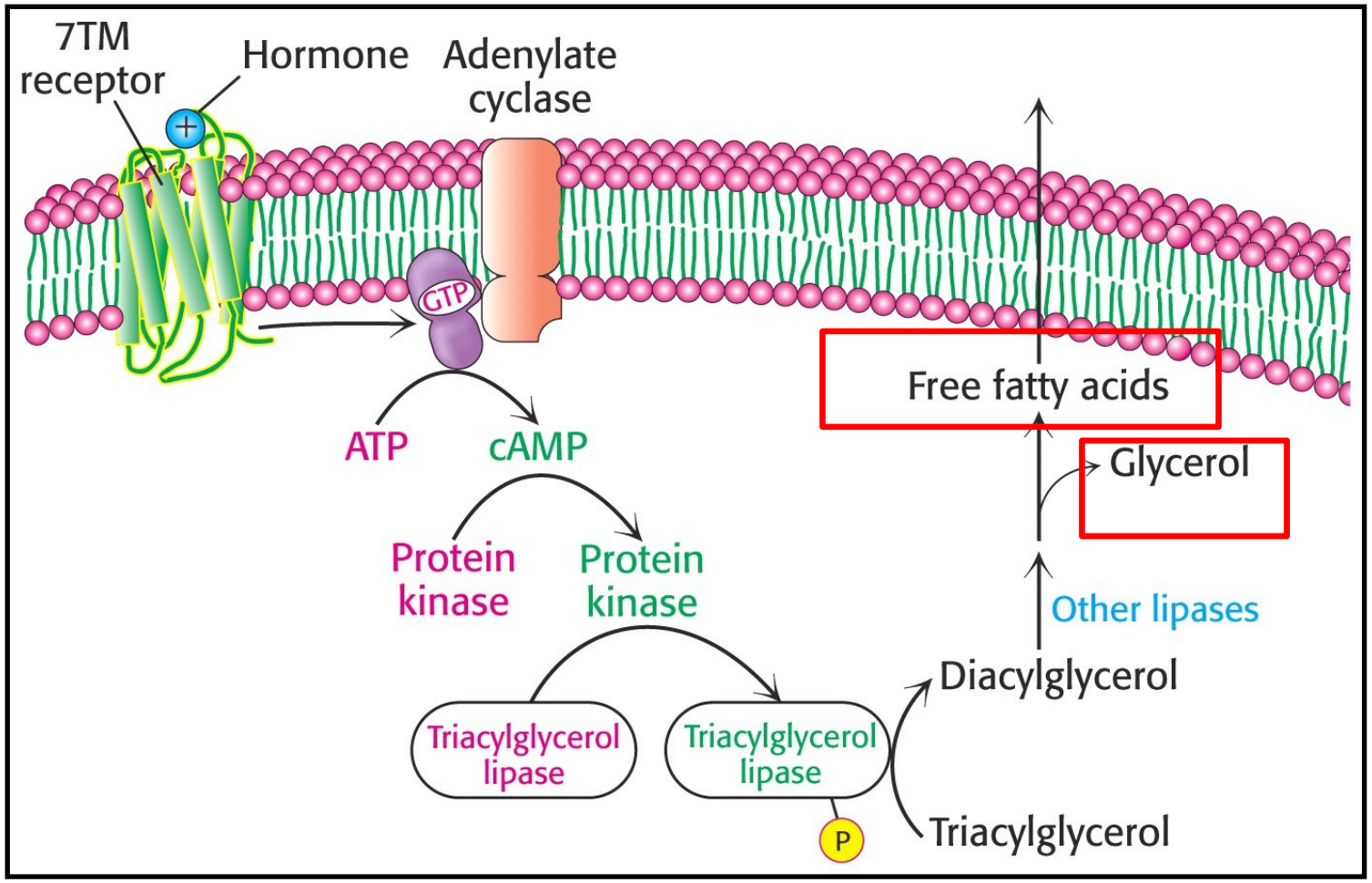
B-lipolysis(Breakdown of TG)

- **Is the breakdown of fats & other lipids by hydrolysis to release FAs.**

- **TG in ATs & liver undergoes hydrolysis by hormone-sensitive TG lipase to form free FAs & glycerol. (hormones such as glucagon and adrenaline)**

- **HSL removes FA from carbon No.1 & or C_3 additional lipase specific for diacyl or monoacylglycerol removes the remaining FAs**

Hormonal regulation of TG degradation in the adipocyte

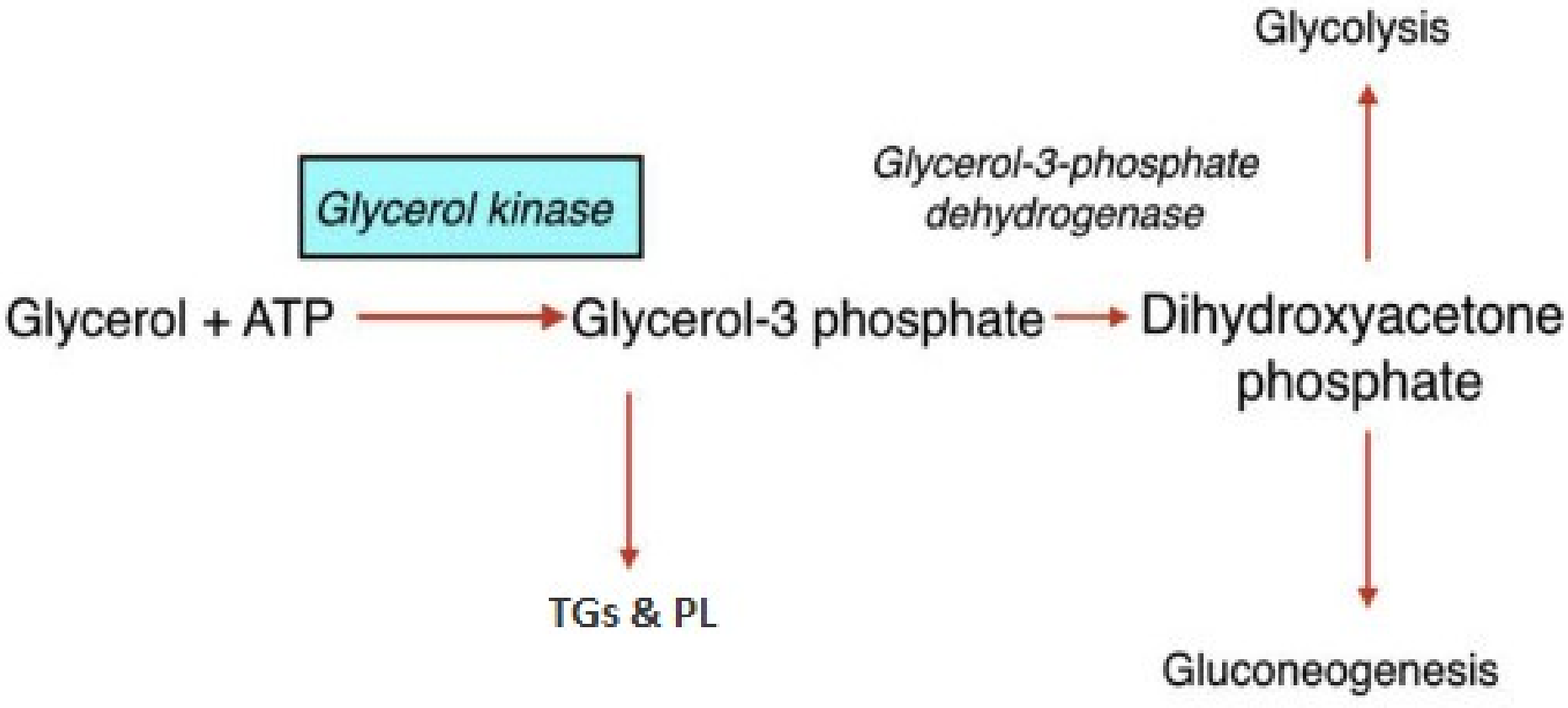


What is the fate of glycerol?

What is the fate of fatty acids?

B.1-The fate of glycerol

The glycerol released during lipolysis is transported through the blood to the liver, where it can be phosphorylated, converted into TGs, or participate in glycolysis and gluconeogenesis.



What is the fate of FAs in the body?

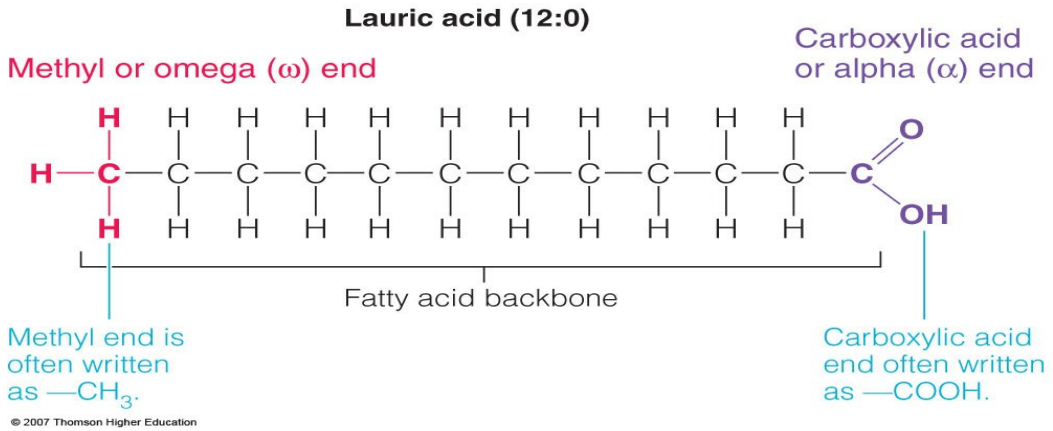
B. 2. The fate of fatty acids

- **Free FAs move through the cell membrane of the adipocyte and immediately bind to albumin in the plasma, then they**
- **Transport to the tissues, where the FA enters the cells, gets activated to their CoA derivatives & are oxidized for energy,**

B.2.1. Fatty acids metabolism:

FAs are water-insoluble long-chain hydrocarbons with one carboxyl group at the end of the chain, which may be

- **Saturated FAs: Do not have a double bond in the chain**
- **Unsaturated FAs: Have one or more double bonds**
- **Branched-chain fatty acids like Phytanic acid**



B.2.2. Naming Systems for FAs:

FAs can be named using several systems:

1.The systematic name (IUPAC):

2.Trivial Names: The names typically derive from a common source of the compound or the source from which it was first isolated.

3.Two Abbreviation Systems:

a. **Alpha (α) Nomenclature (delta (Δ) numbering system**

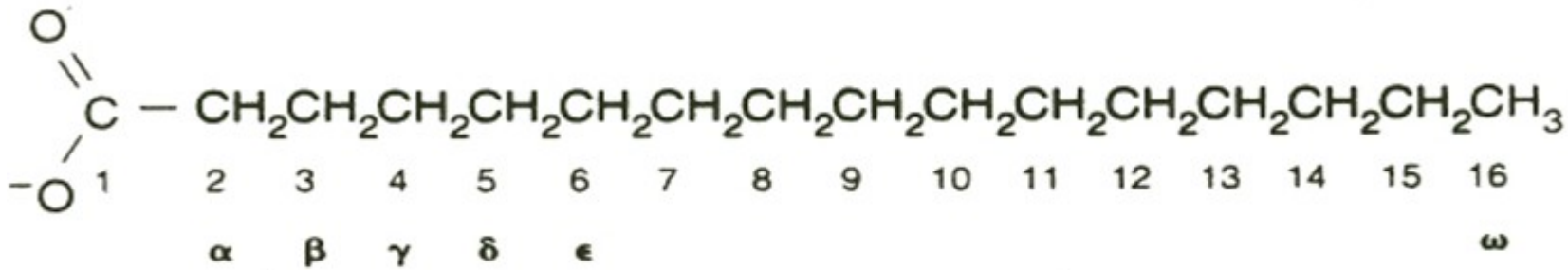
b. **Omega (ω) Nomenclature**

Nomenclature of Fatty Acids

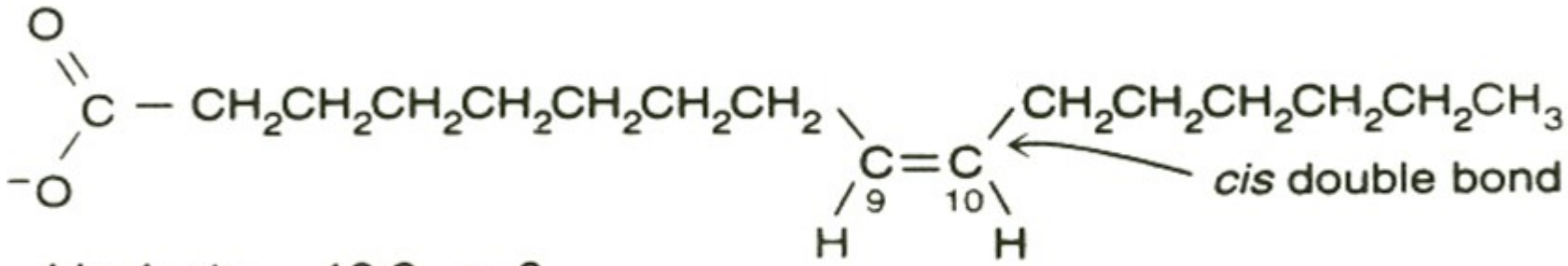
Names		Abbreviations	
trivial	IUPAC	carboxyl-reference	ω - reference
palmitic acid	hexadecanoic acid	16:0	16:0
stearic acid	octadecanoic acid	18:0	18:0
oleic acid	9-octadecenoic acid	18:1 Δ^{9}	18:1 (ω-9)
linoleic acid	9,12-octadecenoic acid	18:2 $\Delta^{9,12}$	18:2 (ω-6)
linolenic acid	9, 12, 15-octadecenoic acid	18:3 $\Delta^{9,12,15}$	18:3 (ω-3)

Note:
Omega nomenclature identifies the position of the first double bond from the methyl (ω) end of the fatty acid.

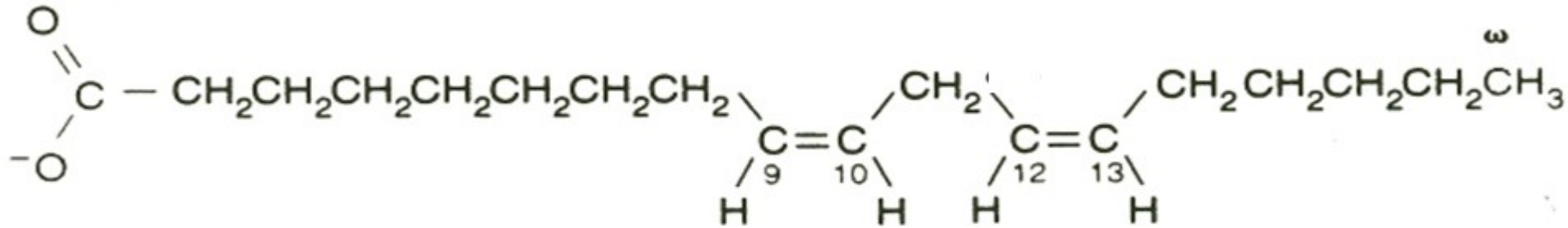
Palmitate 16:0



Palmitoleate 16:1, ω-7



Linoleate 18:2, ω-6



Common name	No. of C & double bond
Saturated FA	
Formic acid	1
Acetic acid	2:0
Butyric acid	4:0
Caproic acid	6:0
Caprylic acid	8:0
Capric acid	10:0
Lauric acid	12:0
Myristic acid	14:0
Palmitic acid	16:0
Stearic acid	18:0
Arachidic acid	20:0
Behenic acid	22:0
Lignocericacid	24:0

Unsaturated FA			
Palmitoleic acid	$16:1\Delta^{9}$		ω-7
Oleic acid	$18:1\ \Delta^{9}$		ω-9
Linoleic acid	$18:2\ \Delta^{9,12}$	Essential FAs	ω-6
Linonlenic acid	$18:3\ \Delta^{9,12,15}$		ω-3
Arachidonic acid	$20:4\ \Delta^{5,8,11,14}$		ω-6
Eicosapentaenoic acid	$20:5\ \Delta^{5,8,11,14,17}$		ω-3

Note: Essential Fas refers to those polyunsaturated fatty acids that must be provided by foods because these cannot be synthesized in the body yet are necessary for health.

What are the different types of dietary fats, and how do they affect our health?

Saturated FFAs

1- Source of energy

2- In the food: Usually accompanied by ch.

3- Enhance the level of T.Ch. & LDL

4- Stability: Autooxidation takes place at higher T

5- Promote obesity & atherosclerosis

Note: Diet must contain a limited amount of saturated FAs to reduce the above disadvantages.

Unsaturated FAs

- Source of energy
- In the food: usually accompanied by ch.
- They resist oxidation
- They are essential nutrients that are important in preventing and managing heart disease
- It is a primary structural component of the human brain, cerebral cortex, skin, & retina.

Note: The cerebral cortex is the thin layer of the brain

What are ω-3 FAs (Fish oil)?

Omega-3 FAs (Fish oil) have significant health benefits and may contribute to improving overall well-being and health.

They cannot be synthesized by the body and must be obtained from the diet may help:

- **Improve Lipid Profile**
- **Inflammation Reduction**
- **Cardiovascular Health**
- **Brain Function**
- **Eye Health**
- **Joint Health**

Example: **Eicosapentaenoic acid, Docosahexaenoic acid & linolenic acid**

Omega-6 FAs

ω-6 FAs also play important roles in the body and have various potential health benefits.

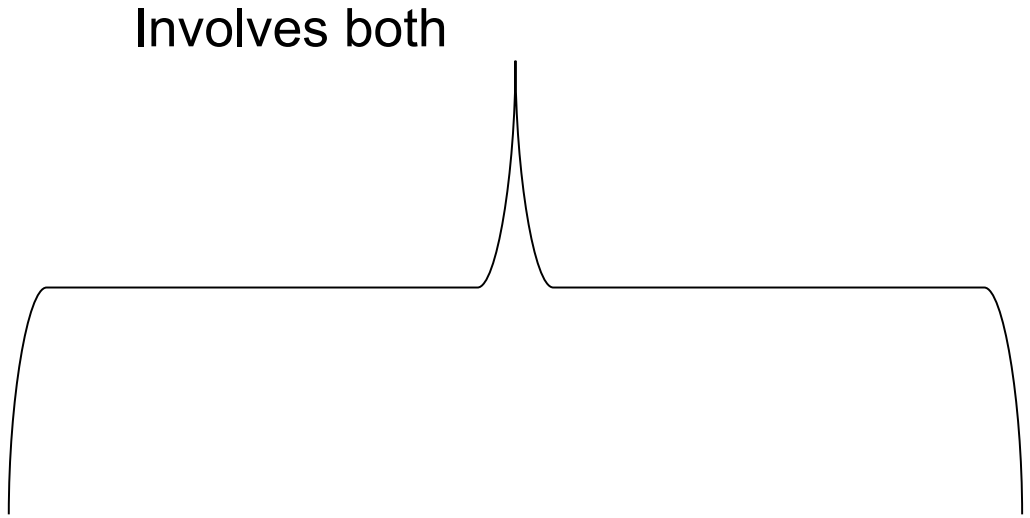
Here are some reasons why ω-6 FAs are helpful:

- **Cell Structure and Function**
- **Inflammatory Response**
- **Skin Health**
- **Brain Function**
- **Hormone Regulation**
- **Heart Health**

Omega-3 FAs & Omega-6 FAs

- **Lipid Metabolism: Helps regulate cholesterol levels by raising HDL and lowering LDL and triglycerides.**
- **Cardiovascular Health: Contribute to improved vascular function, reduced clot formation, and modulation of inflammatory processes, lowering CVD risk.**
- **Brain Function: Supports cognitive function, memory, and mood regulation.**
- **Skin Health: Maintain skin barrier function and hydration.**
- **Hormone Regulation: Precursors for eicosanoids, which regulate processes such as blood pressure, immune response, and inflammation.**
- **Joint Health: May reduce stiffness and pain in arthritis through effects on inflammatory mediators.**

B.2.4. Fatty acids Metabolism



Biosynthesis

Anabolism

Breakdown

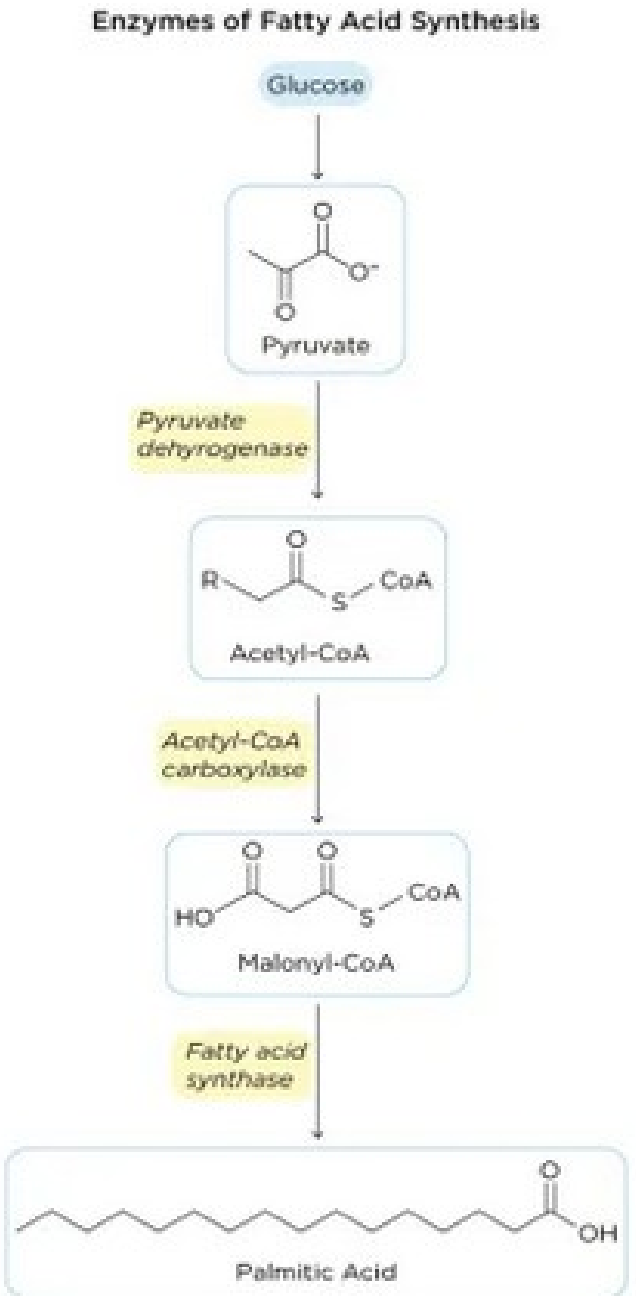
Catabolism

What are the essential components & conditions needed fornon-essentiall fatty acid biosynthesis?

B.2.4.1. Anabolism of FAs

General Features

- ✓ **Glucose is the primary substrate for lipogenesis.**
- ✓ **The major site of FA synthesis is in the liver**
- ✓ **The enzymes that synthesize FA are localized in the cytosol & are completely different from the mitochondrial enzymes that catalyze FA degradation**
- ✓ **The major pathway for FA Biosynthesis involves the polymerization of 2- carbon units derived from acetyl-CoA.**
- ✓ **NADPH, Mn^{+2}, CO_2 & ATP are required for FA biosynthesis.**



Notes:

1. **Essential FAs must be obtained from the diet because there is no human enzyme that can introduce a double bond beyond the 9th carbon atom,**
2. **The major product of FA Synthesis is Palmitate (16C).**

3. Synthesis of unsaturated FA:

The liver & AT contain the necessary enzymes for converting

- ✓ **Palmitic 16:0 to Palmitoleic acid 16:1Δ^9 &**
- ✓ **Stearic 18:0 to Oleic acid 18:1 Δ^9.**

These enzymes are termed Lipoxygenases or mixed-function Oxygenases (requiring molecular O_2, NADH, & Cyt.b_5.

The oxidation of FAs is carried out in 3 key stages:

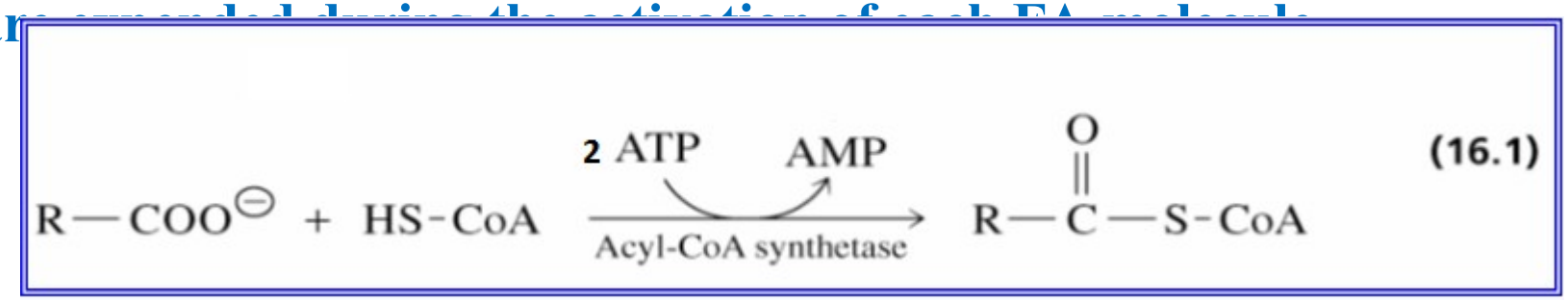
1. **Activation of FFAs**
2. **Transport of FAs into the mitochondria**
3. **Degradation of FAs to acetyl-CoA in the mitochondrial matrix via 3 principal oxidative pathways: α-oxidation, β-oxidation, and ω-oxidation.**

Note: Oxidation of FAs The principal pathways by which FAs are oxidized in the body are: α-Oxidation, β-Oxidation & ω-Oxidation

1- Fatty acid activation

- ✓ **FAs are first activated in the cytoplasm by being converted into fatty acyl-CoA before further metabolism.**
- ✓ **They are then transported into the mitochondria via the carnitine shuttle system.**

Note: 2 ATP ar



2-Transport of acyl-CoA into mitochondria

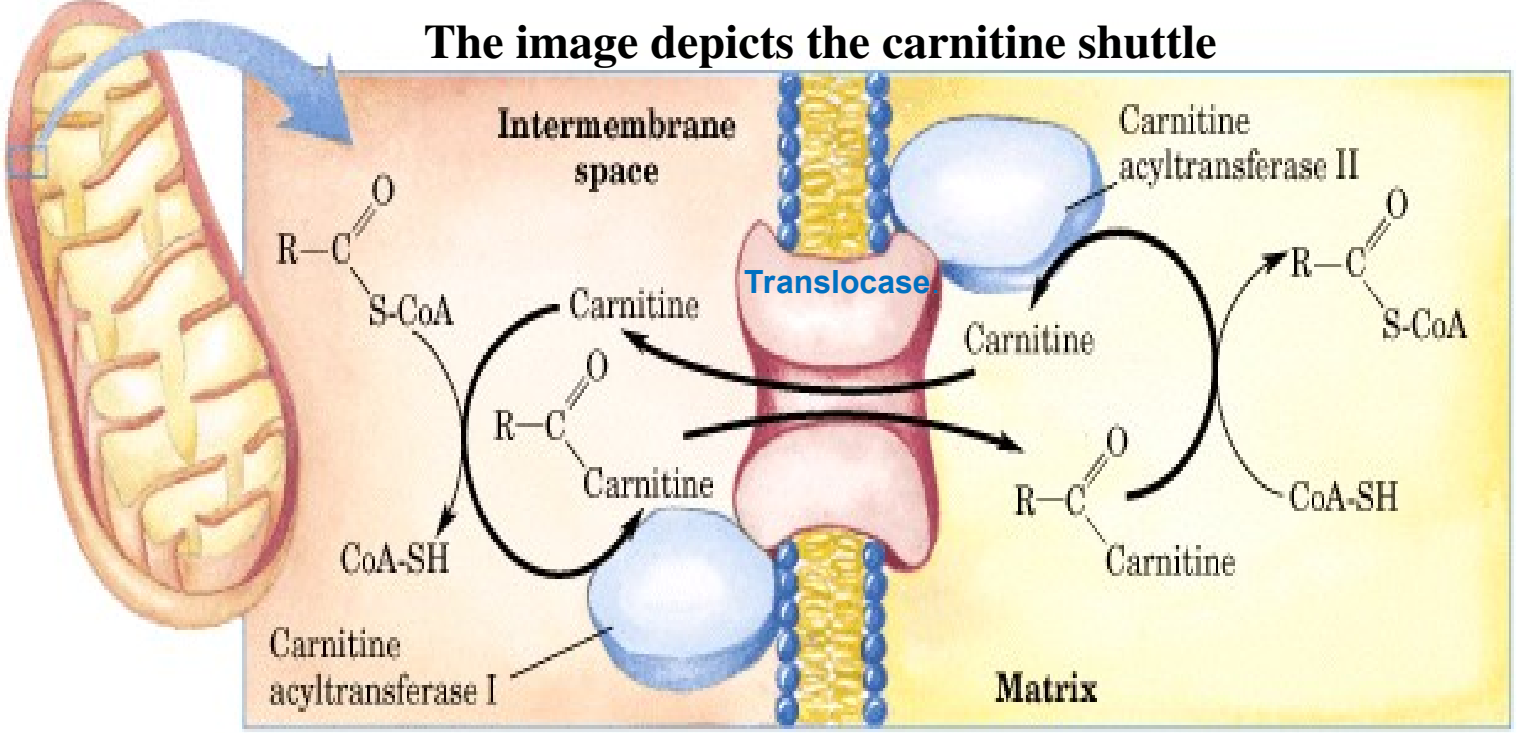
- Most acyl-CoAs are formed outside the mitochondria.
- The oxidation of acyl-CoA is inside the inner membrane.
- The membrane is impermeable to CoA & its derivatives.

How does acetyl-CoA cross the mitochondrial membrane?

Which shuttle transfers acyl-CoA into the mitochondria?

- An efficient shuttle system overcomes this problem called the Carnitine shuttle.
- Carnitine (a natural substance that the body uses to metabolize fats & produce energy) is used as a carrier of the acyl group across the membrane

The image depicts the carnitine shuttle



Carnitine Acyltransferase I Action:

Located on the outer mitochondrial membrane, it transfers the fatty acyl group from CoA to carnitine, forming acyl-carnitine.

Transport Across the Inner Membrane:

Acyl-carnitine is transported into the mito. matrix via **translocase.**

Carnitine Acyltransferase II Action:

Inside the matrix, CAT-II transfers the fatty acyl group back from carnitine to CoA, regenerating fatty acyl-CoA for β-oxidation.

3. Degradation of FAs to acetyl-CoA in the mitochondrial matrix through

- ✓ **α-Oxidation**
- ✓ **β-Oxidation**
- ✓ **ω-Oxidation**

What is β-oxidationn?

β-oxidation

- **The primary pathway for FAs oxidation is known as β-oxidation.**
- **Circulating FAs are absorbed by various tissues, including the liver, heart, kidneys, muscles, lungs, testes, and adipose tissue, where they undergo oxidation.**
- **In cardiac muscle, FAs serve as a crucial energy source for cellular respiration.**

How many stages are there in beta-oxidation?

Beta-oxidation consists of a four-step cycle that repeats until the FA is fully degraded:

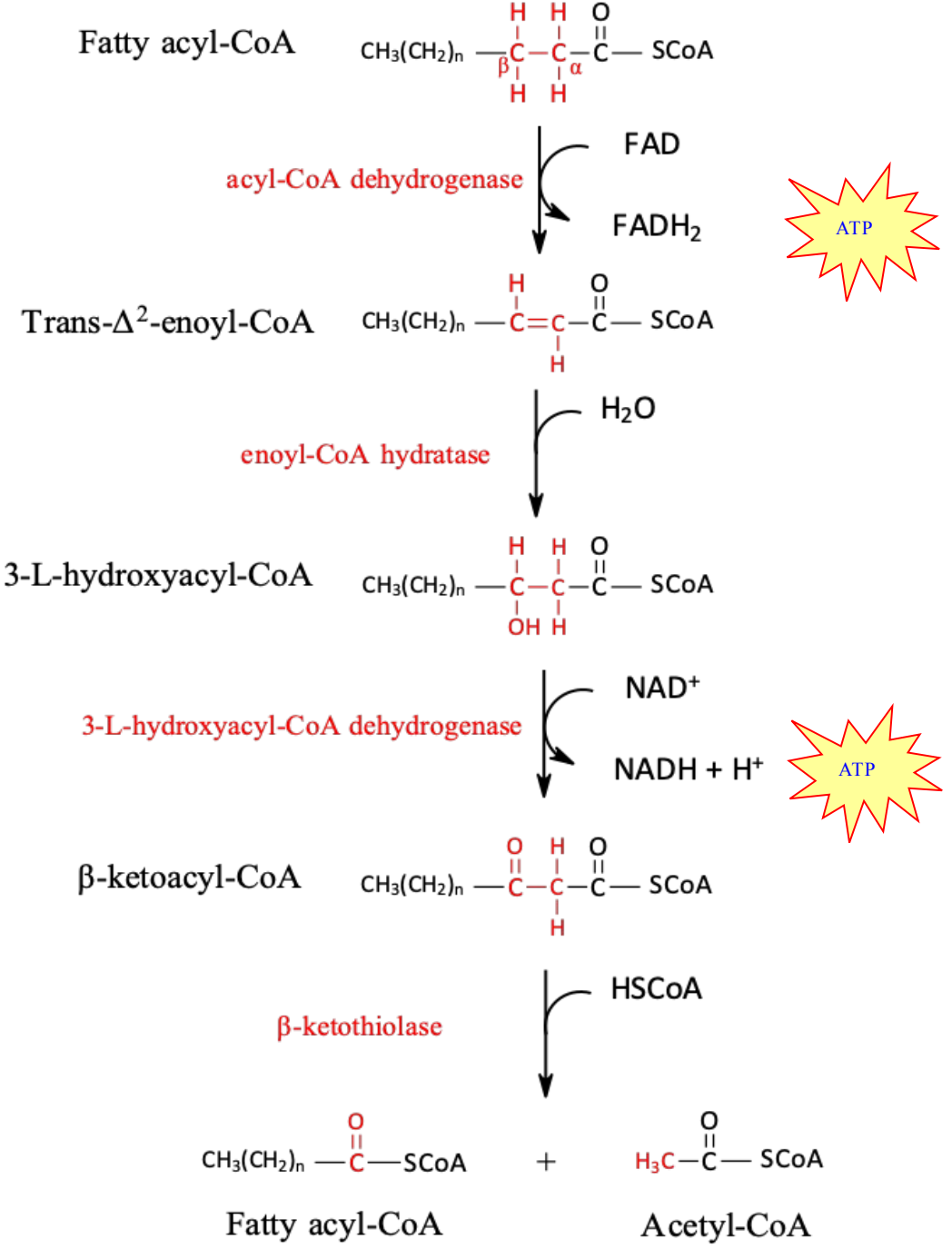
1. **Dehydrogenation linked to FAD**
2. **Hydration**
3. **Re-dehydrogenation linked to NAD^+**
4. **Thiolysis by CoA-SH**

1. **The 1st dehydrogenation of β-oxid. is catalyzed by acyl CoA dehydrogenase to give an enoyl CoA with a trans-double bond between C-2 & C-3.**

2. **The next step is the hydration of the double bond between C-2 & C-3, by enoyl CoA hydratase.**

3. **The 3rd step is dehydrogenation again, catalyzed by L-β-hydroxyl acyl CoA dehydrogenase. The products are ketoacyl CoA and NADH.**

4. **The final step is the cleavage of β-ketoacyl-CoA by the thiolase. The products are acetyl-CoA and acyl CoA shortened by 2 carbon atoms.**



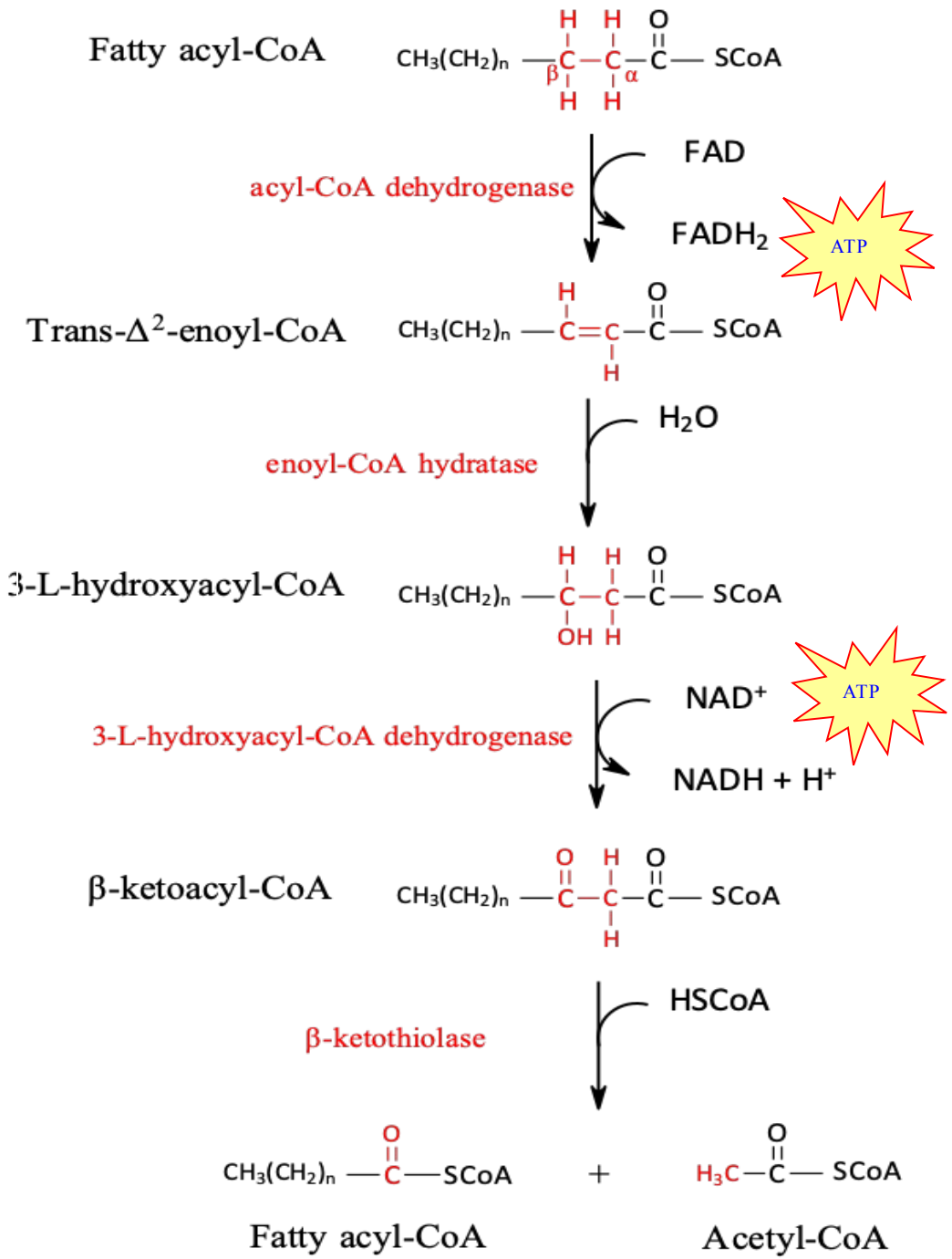
Key Points:

✓**Each cycle shortens the fatty acyl-CoA by 2 carbons.**

✓**Products per cycle:**

- **1 Acetyl-CoA,**
- **1 NADH, and**
- **1 $FADH_2$.**

✓**The shortened acyl-CoA enters another β-oxidation cycle, & this process repeats until the entire FA is broken down into Acetyl-CoA units.**



- ?How do you calculate ATP yield from β-Oxidation
- ?How many ATPs are generated from the oxidation of palmitate

Bioenergetics of FA oxidation

Ex. Palmitic acid **Note: 16:0 C**

1. In each reaction cycle of β-Oxidation, an acyl-CoA is shortened by 2 Cs.

How many cycles (Beta–oxidation) are needed for the complete oxidation of Palmitic acid 16:0 C?

2. The full breakdown of palmitoyl-CoA requires 7 cycles of β-oxidation.
3. After the 7th cycle, the process yields an 8th acetyl-CoA (CH_2-CO-CoA).

ATP Yield from Palmitic Acid (16:0) Oxidation

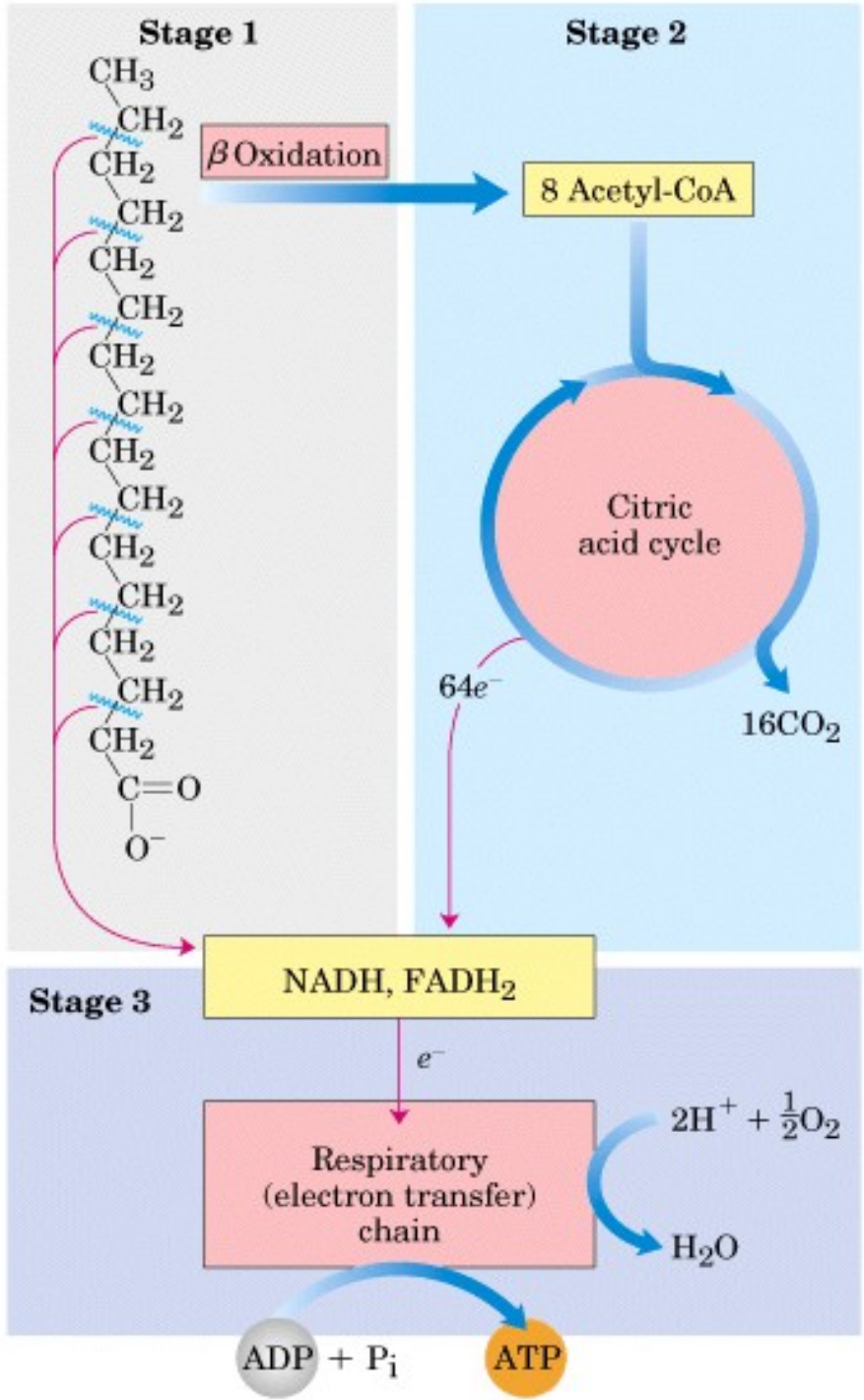
- ✓ **How Many Cycles Are Needed? = 7**

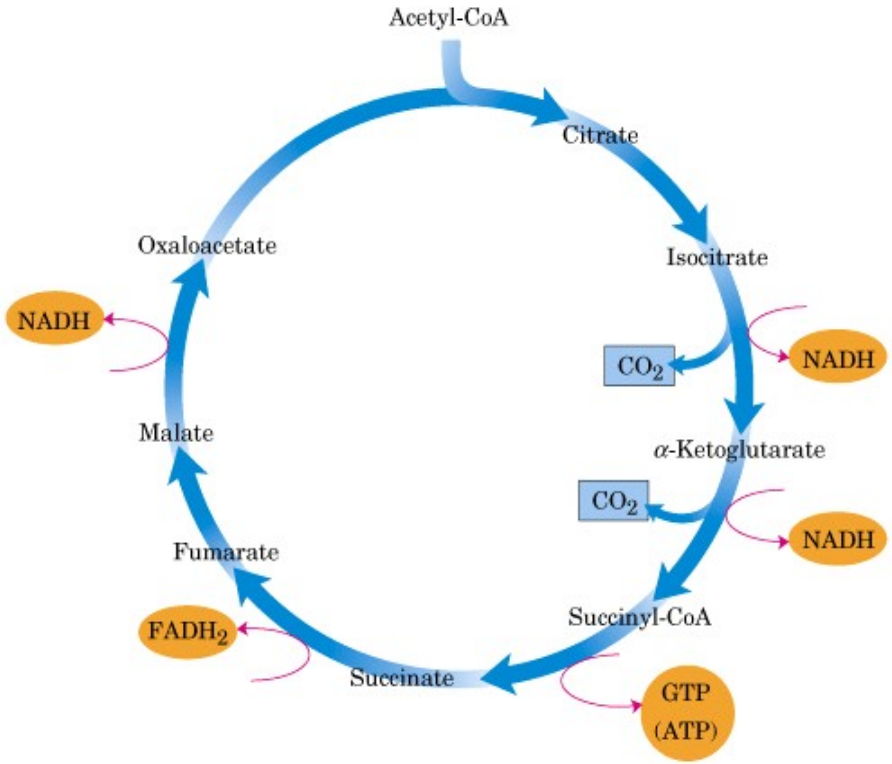
  Number of cycles=(N/2)–1=7 N=No of carbons

- ✓ **Products per β-Oxidation cycle:**
  - **1 Acetyl-CoA,**
  - **1 NADH, (7 NADH =3 ATP*7= 21 ATP) &**
  - **1 $FADH_2$ (7 FADH2 =2 ATP*7= 14 ATP)**
- ✓ **The number of acetyl-CoA produced from Palmitic Acid (16:0) is 8 acetyl-CoA**
- **8 Acetyl-CoA enters the TCA cycle: 8*12 = 96 (12 ATPs are produced for each acetyl-CoA in the TCA cycle)**

Total No. of ATP from 1 molecule of palmitate = 96 + 21 + 14 = 131.

- **But remember, we used 2 ATP to activate palmitate to palmitoyl-CoA. Therefore, the ATP yield is 129.**


Stage 1
Stage 2
Stage 3
β Oxidation
8 Acetyl-CoA
Citric acid cycle
64e⁻
16CO2
NADH, FADH2
e⁻
Respiratory (electron transfer) chain
2H⁺ + ½O2
H2O
ADP + Pi
ATP


Acetyl-CoA
Citrate
Isocitrate
NADH
CO2
NADH
α-Ketoglutarate
CO2
NADH
Succinyl-CoA
GTP (ATP)
Succinate
FADH2
Fumarate
Malate
Oxaloacetate
NADH

How do odd-chain FAs undergo oxidation?

What is the mechanism behind the oxidation of odd-chain FAs?

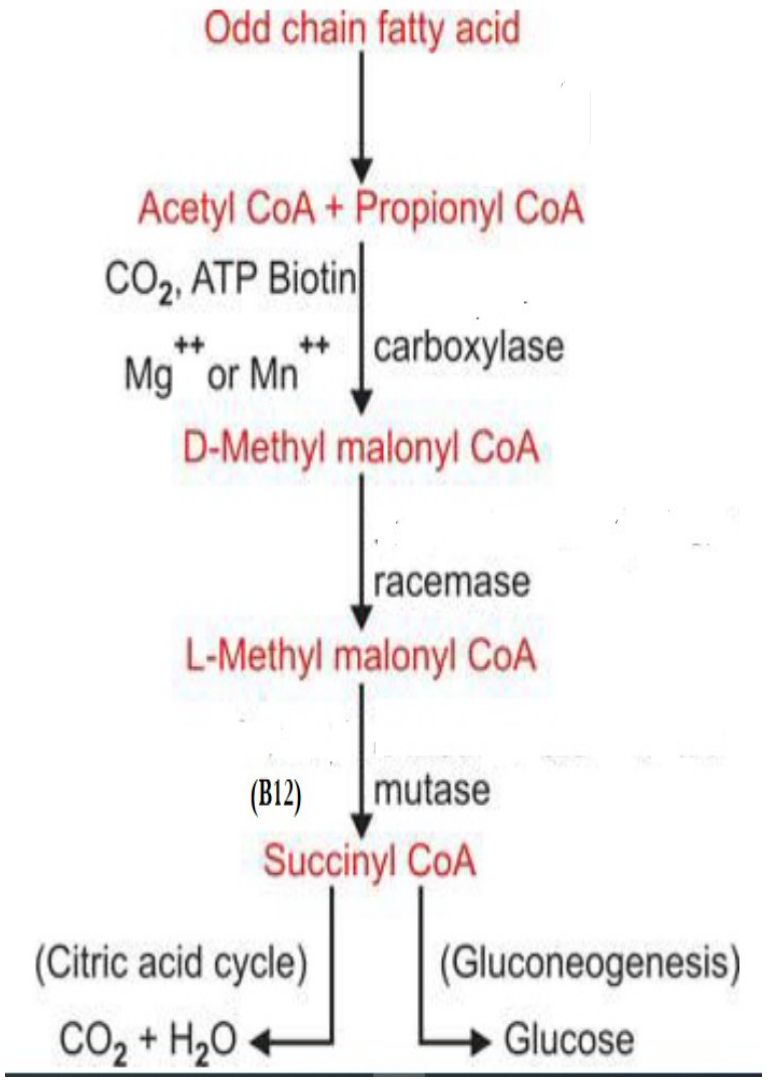
Oxidation of odd-chain FAs

- ✓ **The oxidation of FAs with an odd number of carbons proceeds exactly as the β-oxidation of FAs. However, the final product is a molecule of Propionyl CoA.**

- ✓ **Propionyl-CoA further undergoes carboxylation, molecular rearrangement, & conversion to succinyl-CoA,**

- ✓ **The succinyl CoA can be metabolized via the TCA cycle, of which it is an intermediate, or it could be the substrate for heme synthesis and or gluconeogenesis.**

Metabolism of propionyl CoA



What is the mechanism behind the oxidation of **unsaturated** fatty acids?

Oxidation of unsaturated FAs

Note :

A.The oxidation of unsaturated FAs requires the involvement of 2 additional enzymes:

- ✓**Isomerase &**
- ✓**Epimerase.**

A.Notably, the presence of a double bond in an FA leads to a reduction in ATP yield during oxidation by two units(-2ATP).

What is ω- Oxidation?

ω- Oxidation:

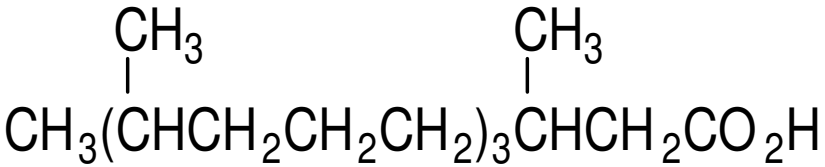
- **There is an alternate pathway to β-oxidation called ω-oxidation. Instead of involving the β-C, this pathway involves the oxidation of the ω-C by hydroxylase enzymes involving**
  - ✓ **Cyt P450 &**
  - ✓ **Mono-oxygenase.**

- **ω-Oxidation is typically a minor catabolic pathway for medium-chain FAs, but it becomes more critical when β-oxidation is impaired.**

What is α - Oxidation?

α-Oxidation

- ✓ **This metabolic process is responsible for breaking down branched-chain FAs by removing a single carbon atom from the carboxyl end of the molecule.**
- ✓ **In humans, this process takes place exclusively within peroxisomes and is essential for the degradation of phytanic acid, a fatty acid derived from chlorophyll & found in certain foods.**



Phytanic Acid is a branched-chain FA, that can't be oxidized by β-oxidation. due to its β-methyl branch (β-C atom blocked by methyl group)

How does the body metabolize cholesterol?

Cholesterol Metabolism

- ✓ **Cholesterol is a waxy, fat-like substance found in all the cells of the body**
- ✓ **The total amount of Ch. in the human body is about 2g/kg body weight.**

It is found in high concentrations in:

- ➢ **Adrenal gland** **100mg/g**
- ➢ **Brain** **20mg/g**
- ➢ **Liver** **3mg/g**
- ➢ **Skeletal muscle** **1mg/g**

- **The majority of the body's cholesterol is produced through biosynthesis, approximately 1 gram per day, while the diet provides only around 0.3 grams per day.**

- **Cholesterol is a product of animal metabolism and can be found in foods of animal origin, such as meat, liver, brain, and egg yolk.**

Cholesterol biosynthesis:

- **De novo biosynthesis of cholesterol occurs in almost all cells, but this process is more predominant in the**
  - **Liver,**
  - **Intestine,**
  - **Adrenal cortex, &**
  - **Reproductive tissues, including ovaries, testis, & placenta.**

- **The majority of endogenous cholesterol (about 70-80%) is formed in the liver, while only 10% is produced in the intestine**

- **The major site:** Cytosol & endoplasmic reticulum
- **The raw material:** Acetyl CoA
- **The cofactors:** ATP, NADPH & NADH

Cholesterol Biosynthesis is

- **26 steps**
- **3 stages**

A-Formation of HMG-CoA:

- **3 moles of acetyl-CoA are used to synthesize 3-hydroxyl-3-methylglutaryl-CoA (HMG-CoA)**

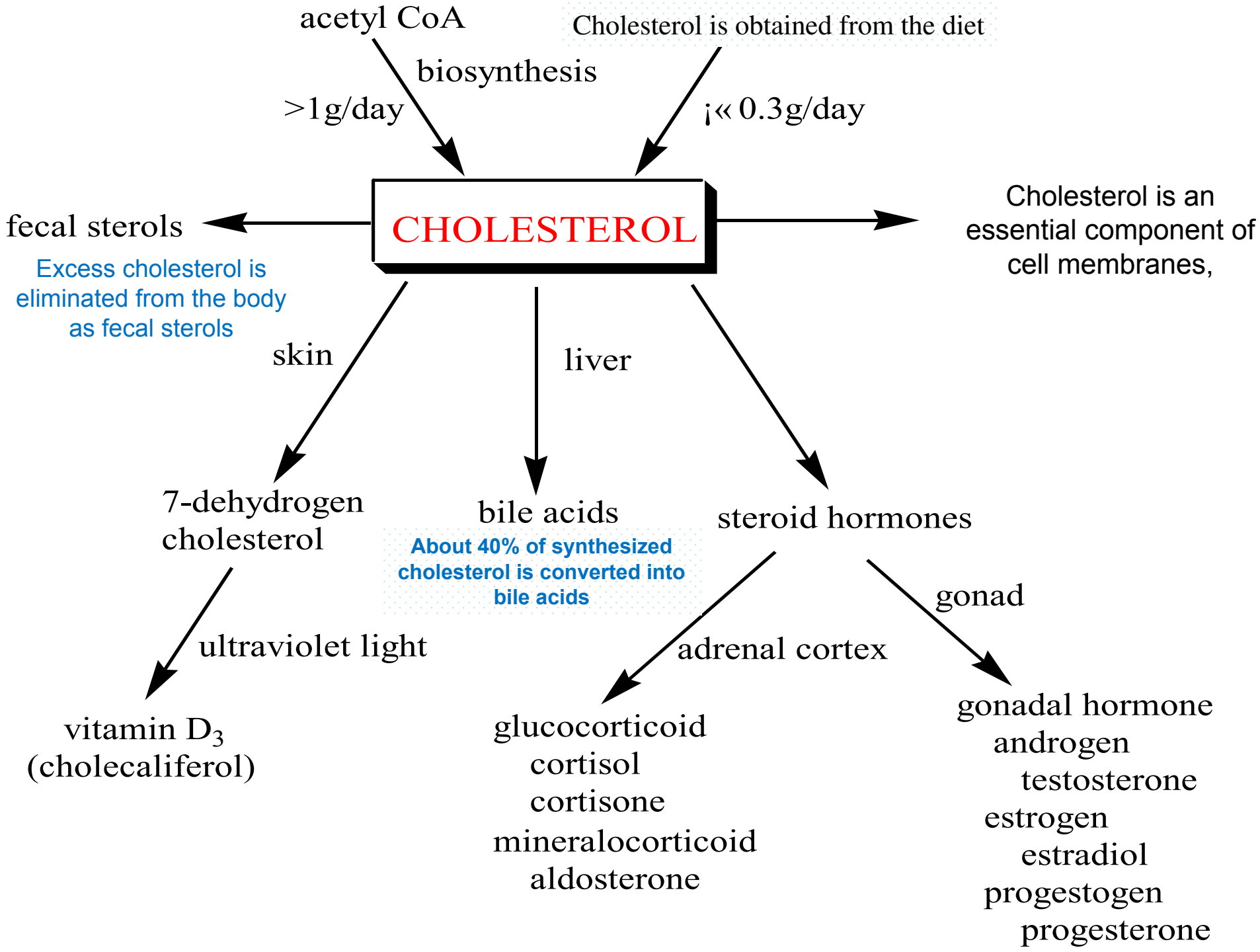
B-Conversion of HMG-CoA to Squalene

- **The rate-limiting step of chol. Biosynthesis is the conversion of HMG-CoA to mevalonic acid, catalyzed by the enzyme HMG-CoA reductase.**

C-Formation of cholesterol from squalene

How does the body metabolize cholesterol?

Metabolic conversion of cholesterol


acetyl CoA
biosynthesis
>1g/day
Cholesterol is obtained from the diet
¡« 0.3g/day
CHOLESTEROL
fecal sterols
Excess cholesterol is eliminated from the body as fecal sterols
Cholesterol is an essential component of cell membranes,
skin
liver
7-dehydrogen cholesterol
bile acids
About 40% of synthesized cholesterol is converted into bile acids
steroid hormones
ultraviolet light
adrenal cortex
gonad
vitamin D3 (cholecaliferol)
glucocorticoid
cortisol
cortisone
mineralocorticoid
aldosterone
gonadal hormone
androgen
testosterone
estrogen
estradiol
progestogen
progesterone

What are the regulatory pathways involved in cholesterol metabolism?

How is cholesterol metabolism regulated?

Regulation of cholesterol metabolism

1. **HMG-CoA reductase is a key enzyme that catalyzes the conversion of HMG-CoA to mevalonate. This enzyme can be inhibited strongly by cholesterol.**

2. **Dietary Influence: When the diet contains 2% cholesterol, the body's endogenous cholesterol production decreases by 10-30%. However, increasing dietary cholesterol further does not completely suppress endogenous cholesterol synthesis.**

What are ketone bodies?

Formation & utilization of ketone bodies

- **The last product of FAs in β-oxi. is acetyl CoA, which in many tissues, except the liver, can be completely oxidized to CO_2 & water. However, in liver cells, a significant amount of acetyl-CoA is converted to so-called ketone bodies.**

Which include 3 substances:

- ❖ **Acetoacetate**
- ❖ **β-Hydroxybutyrate**
- ❖ **Acetone**

Ketone bodies are water-soluble; their con. in the blood is less than 1 mg/dl .

- **In the early stages of fasting, the heart and skeletal muscles utilize ketone bodies as an energy source, helping to preserve glucose for the central nervous system.**
- **With more prolonged starvation, the brain starts using more ketone bodies for energy to spare glucose consumption**

➢ **These molecules can provide energy for the body. Unlike fatty acids, ketones can cross the blood-brain barrier & provide energy for the brain in the absence of glucose.**

➢ **Ketosis is a metabolic state where ketones become an important source of energy for the body & brain.**

Clinical disorders associated with lipid and lipoprotein metabolism

A- Refsum's Disease

- A rare inherited disorder in which phytanic acid accumulates in tissues.
- Possibly due to a defect or deficiency of the α-hydroxylase, which causes

  - Nerve & retinal damage
  - Bone & skin changes

B-Xanthomatosis:

Is the accumulation of lipids, usually, cholesterol in subcutaneous tissue with large foam cells. It appears in hyper-lipoproteinemia, diabetes mellitus & Von –Gierke's disease





C-Essential Fatty Acid Deficiency

It is a rare condition, occurring most often in infants fed diets deficient in EFAs. Signs include

- **Irritated & flaky skin (Skin lesion)**
- **GI problems**
- **Compromised immune system**
- **Slow growth in children**
- **kidney damage**
- **Decreased resistance to stress**

D-Atherosclerosis

- ✓ **Is a progressive disease characterized by the buildup of plaque inside the arteries.**
- ✓ **Plaque consists of fatty substances, cholesterol, cellular waste products, Ca, and fibrin (a clotting material). Over time, this buildup narrows the arteries, reducing blood flow to vital organs and tissues.**
- ✓ **Is the first step in the development of CVD,**

E- L-Carnitine deficiency

- **Carnitine deficiency is a condition where the nutrient can't reach the body's cells**
- **Is an AA. that is naturally produced in the body.**
- **It helps the body produce energy.**
- **It is important for heart & brain function, muscle movement, & many other body processes.**

Carnitine deficiencies can cause

- **Poor muscle tone**
- **Muscle weakness**
- **Brain dysfunction**
- **Heart dysfunction**

Hyper lipoproteinemia

Disorders	Defect
Type I **Familial Chylomicronemia**	**a) Deficiency of LPL** **b) Production of abnormal LPL** **c) Apo-C deficiency**
.Type II-Familial Ch	**LDL receptor defect**
Type III **(familial Dysbetalipoproteinemia)**	**Apo-E abnormality**
Type IV -Familial TG.	**Elevated production of VLDL**
Type V	**Elevated chylomicrons & VLDLs due to an unknown cause**
Familial hyper alpha lipoproteinemia	**Increased level of HDLs**
Familial LCAT deficiency	**Absence of LCAT leads to the inability of HDLs to take up cholesterol (reverse cholesterol transport)**

Hypolipoproeinemia

Disorders	Defect
β-lipoproteinemia	No chylomicrons, VLDLs or LDLs due to defect in apo-B expression
Familial hypo β-lipoproteinemia	Apo B gene mutations
Familial α-lipoprotein deficiency	Apo-A_I & C_{III}